

Tuffile and Pinho (1970) compared a number of methods for the determination of K_La values in viscous streptomycete fermentations. The techniques used were static gassing-out, dynamic gassing-out and the oxygen-balance method. Tuffile and Pinho did not make it clear whether non-respiring mycelium was present during their static gassing-out procedure, but from their results it would appear that it was present in the vessel. Thus, the rheology of the fermenter contents would appear to have been similar for the different determinations. The K_La values, determined by the different techniques, for a 300-dm³ fermenter containing a 90-hour culture of *Streptomyces aureofaciens* are shown in Table 9.4.

From Table 9.4 it may be seen that the K_La values for the two gassing-out techniques were very similar but there was a considerable difference between the oxygen-uptake rates and the K_La s determined by the dynamic method and the balance method. Tuffile and Pinho (1970) claimed that the low oxygen-uptake rate determined by the dynamic method was due to air bubbles remaining in suspension in the mash during the dynamic gassing-out period. Thus, the decline in oxygen concentration after the cessation of aeration was not a measure of the oxygen-uptake rate but the difference between oxygen uptake and the transfer of oxygen from entrapped bubbles. It was demonstrated that a large number of bubbles remained suspended in the medium 15 minutes after aeration had been stopped. The use of the low oxygen-uptake rate in the calculation of the K_La would result in an artificially low K_La being determined. Heijnen *et al.* (1980) also observed anomalies in determining K_La values in viscous systems due to the presence of very small bubbles having a much longer residence time than the more abundant large bubbles in the vessel.

Overall, it would appear that the balance method is the most desirable technique to use and the extra cost of the monitoring equipment involved should be a worthwhile investment.

Before considering the factors which may affect the

K_La of a fermenter it is necessary to consider the behaviour of fluids in agitated systems.

FLUID RHEOLOGY

Fluids may be described as Newtonian or non-Newtonian depending on whether their rheology (flow) characteristics obey Newton's law of viscous flow. Consider a fluid contained between two parallel plates, area A and distance x apart. If the lower plate is moved in one direction at a constant velocity, the fluid adjacent to the moving plate will move in the same direction and impart some of its momentum to the 'layer' of liquid directly above it causing it, also, to move in the same direction at a slightly lower velocity. Newton's law of viscous flow states that the viscous force, F , opposing motion at the interface between the two liquid layers, flowing with a velocity gradient of dv/dx , is given by the equation:

$$F = \mu A(dv/dx) \quad (9.7)$$

where μ is the fluid viscosity, which may be considered as the resistance of the fluid to flow.

Equation (9.7) may be written as:

$$F/A = \mu(dv/dx)$$

F/A is termed the shear stress (τ) and is the applied force per unit area, dv/dx is termed the shear rate (γ) and is the velocity gradient. Thus:

$$\tau = \mu\gamma \quad (9.8)$$

Equation (9.8) conforms to the general relationship:

$$\tau = K\gamma^n \quad (9.9)$$

where K is the consistency coefficient and n is the flow behaviour index or power law index.

For a Newtonian fluid n is 1 and the consistency coefficient is the viscosity which is the ratio of shear stress to shear rate. Thus, a plot of shear stress against

TABLE 9.4. K_La values for a 300-dm³ fermenter containing a 90-h culture of *S. aureofaciens* (Tuffile and Pinho, 1970)

Method of K_La determination	Measured oxygen uptake rate (mmoles dm ⁻³ h ⁻¹)	K_La (h ⁻¹)
Static gassing out	—	58.2
Dynamic gassing out	6.6	58.2
Oxygen balance	20.1	108.0

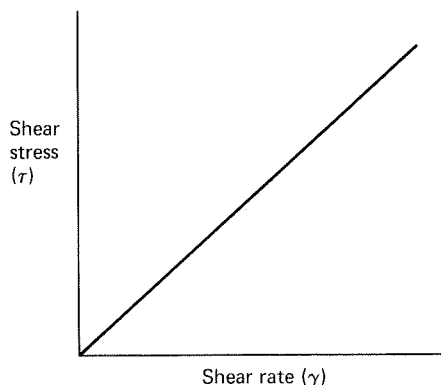


FIG. 9.10. A rheogram of a Newtonian fluid.

shear rate, for a Newtonian fluid, would produce a straight line, the slope of which would equal the viscosity. Such a plot is termed a rheogram (as shown in Fig. 9.10).

Thus, a Newtonian liquid has a constant viscosity regardless of shear, so that the viscosity of a Newtonian fermentation broth will not vary with agitation rate. However, a non-Newtonian liquid does not obey Newton's law of viscous flow and does not have a constant viscosity. The value for n (equation (9.9)) of such a fluid deviates from 1 and its behaviour is said to follow a power law model. Thus, the viscosity of a non-Newtonian fermentation broth will vary with agitation rate and is described as an apparent viscosity (μ_a). A plot of shear stress against shear rate for a non-Newtonian liquid will deviate from the relationship depicted in Fig. 9.10, depending on the nature of the liquid. Several types of non-Newtonian liquids are recognized and typical rheograms of types important in the study of culture fluids are given in Fig. 9.11, and their characteristics are discussed below.

Bingham plastic rheology

Bingham plastics are similar to Newtonian liquids apart from the fact that shear rate will not increase until a threshold shear stress is exceeded. The threshold shear stress is termed the yield stress or yield value, τ_0 . A linear relationship of shear stress to shear rate is given once the yield stress is exceeded and the slope of this line is termed the coefficient of rigidity or the plastic viscosity. Thus, the flow of a Bingham plastic is described by the equation:

$$\tau = \tau_0 + n\gamma$$

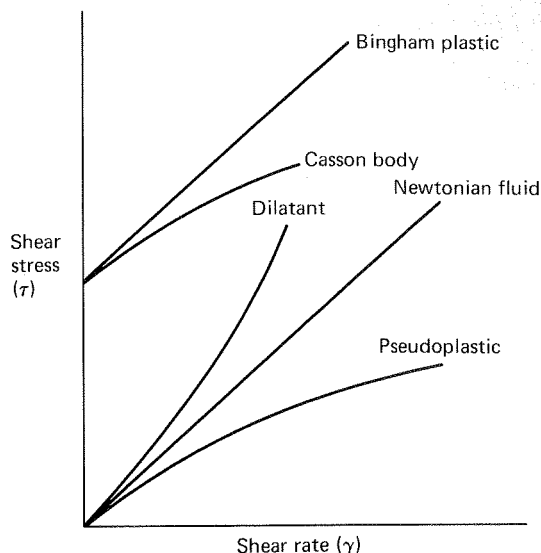


FIG. 9.11. Rheogram of fluids of different properties.

where n is the coefficient of rigidity and τ_0 is the yield stress.

There have been some claims of mycelial fermentation broths displaying Bingham plastic characteristics (Table 9.5). Everyday examples of these fluids include toothpaste and clay.

Pseudoplastic rheology

The apparent viscosity of a pseudoplastic liquid decreases with increasing shear rate. Most polymer solutions behave as pseudoplastics. The decrease in apparent viscosity is explained by the long chain molecules tending to align with each other at high shear rates resulting in easier flow. The flow of a pseudoplastic liquid may be described by the power law model, equation (9.9), i.e.:

$$\tau = K(\gamma)^n$$

K has the same units as viscosity and may be taken as the apparent viscosity. The flow-behaviour index is less than unity for a pseudoplastic liquid, the smaller the value of n , the greater the flow characteristics of the liquid deviate from those of a Newtonian fluid. Equation (9.9) may be converted to the logarithmic form as:

$$\log \tau = \log K + n \log \gamma \quad (9.10)$$

Thus, a plot of log shear stress against log shear rate will produce a straight line, the slope of which will

TABLE 9.5. Some examples of the rheological nature of fermentation broths

Organism	Rheological type	Reference
<i>Penicillium chrysogenum</i>	Bingham plastic	Deindoerfer and Gaden (1955)
<i>Streptomyces kanamyceticus</i>	Bingham plastic	Sato (1961)
<i>Penicillium chrysogenum</i>	Pseudoplastic	Deindoerfer and West (1960)
<i>Endomyces</i> sp.	Pseudoplastic	Taguchi <i>et al.</i> (1968)
<i>Penicillium chrysogenum</i>	Casson body	Roels <i>et al.</i> (1974)

equal the flow-behaviour index and the intercept on the shear stress axis will be equal to the logarithm of the consistency coefficient.

Many workers have demonstrated that mycelial fermentation broths display pseudoplastic properties as shown in Table 9.5.

Dilatant rheology

The apparent viscosity of a dilatant liquid increases with increasing shear rate. The flow of a dilatant liquid may also be described by equation (9.9) but in this case the value of the flow-behaviour index is greater than 1, the greater the value the greater the flow characteristics deviate from those of a Newtonian fluid. Thus, the values of K and n may be obtained from a plot of log shear stress against log shear rate. Fortunately this type of behaviour is not exhibited by fermentation broths — an everyday example is liquid cement slurry.

Casson body rheology

Casson (1959) described a type of non-Newtonian fluid, termed a Casson body, which behaved as a pseudoplastic in that the apparent viscosity decreased with increasing shear rate but displayed a yield stress and, therefore, also resembled a Bingham plastic. The flow characteristics of a Casson body may be described by the following equation:

$$\sqrt{\tau} = \sqrt{\tau_0} + K_c \sqrt{\dot{\gamma}} \quad (9.11)$$

where K_c is the Casson viscosity.

A plot of $\sqrt{\tau}$ against $\sqrt{\dot{\gamma}}$ will give a straight line, the slope of which will equal the Casson viscosity and the intercept of the $\sqrt{\tau}$ axis will equal $\sqrt{\tau_0}$.

Roels *et al.* (1974) claimed that the rheology of a penicillin broth could be best described in terms of a Casson body.

Therefore, to determine the rheological nature of a

fluid it is necessary to construct a rheogram which requires the use of a viscometer which is accurate over a wide range of shear rates. Furthermore, the testing of mycelial suspensions may present special difficulties. These problems have been considered in detail by Van't Riet and Tramper (1991), whose book should be consulted for methods of assessing the rheological properties of mycelial fluids.

FACTORS AFFECTING $K_L a$ VALUES IN FERMENTATION VESSELS

A number of factors have been demonstrated to affect the $K_L a$ value achieved in a fermentation vessel. Such factors include the air-flow rate employed, the degree of agitation, the rheological properties of the culture broth and the presence of antifoam agents. If the scale of operation of a fermentation is increased (so-called 'scale-up') it is important that the optimum $K_L a$ found on the small scale is employed in the larger scale fermentation. The same $K_L a$ value may be achieved in different sized vessels by adjusting the operational conditions on the larger scale and measuring the $K_L a$ obtained. However, quantification of the relationship between operating variables and $K_L a$ should enable the prediction of conditions necessary to achieve a particular $K_L a$ value. Thus, such relationships should be of considerable value in scaling-up a fermentation and in fermenter design.

The effect of air-flow rate on $K_L a$

MECHANICALLY AGITATED REACTORS

The effect of air flow rate on $K_L a$ values in conventional agitated systems is illustrated in Fig. 9.12. The quantitative relationships between aeration and $K_L a$ for agitated vessels are considered in the subsequent

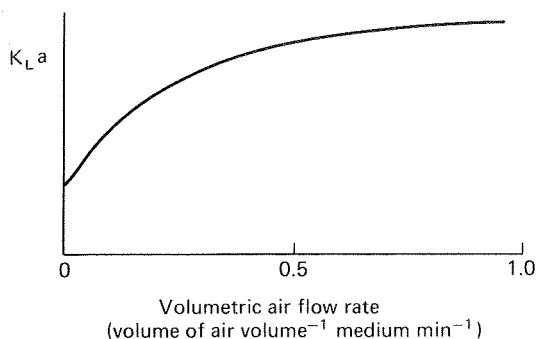


FIG. 9.12. The effect of air-flow rate on the $K_L a$ of an agitated, aerated vessel.

section on power consumption. The air-flow rate employed rarely falls outside the range of 0.5–1.5 volumes of air per volume of medium per minute and this tends to be maintained constant on scale-up. If the impeller is unable to disperse the incoming air then extremely low oxygen transfer rates may be achieved due to the impeller becoming 'flooded'. Flooding is the phenomenon where the air-flow dominates the flow pattern and is due to an inappropriate combination of air flow rate and speed of agitation (see also Chapter 7). Nienow *et al.* (1977) categorized the different flow patterns produced by a disc turbine that occur under a range of aeration and agitation conditions (Fig. 9.13) and these have been discussed further by Van't Riet and Tramper (1991). Figure 9.13 A shows the flow profile of a non-aerated vessel and Figs 9.13 B to F the profiles with increasing air flow rate. As air-flow rate increases the flow profile changes from one dominated by agitation (Fig. 9.13 B) to one dominated by air flow (Figs

9.13 D to F) until finally the air flow rate is such that the air escapes without being distributed by the agitator (Fig. 9.13 F). Different workers have used different criteria to define the onset of flooding with Nienow *et al.* (1977) claiming it to be represented by Fig. 9.13 D whereas Biesecker (1972) suggested Fig. 9.13 F. However, the desired pattern is represented by Fig. 9.13 C.

Several workers have produced empirical quantitative descriptions of flooding systems which may assist in avoiding the phenomenon:

- (i) Westerterp *et al.* (1963) calculated that the minimum impeller tip speed to avoid flooding should be between 1.5 and 2.5 m second^{-1} .
- (ii) Biesecker (1972) claimed that flooding occurs when the energy dissipated by the air flow is greater than that dissipated by the agitator. Van't Riet and Tramper (1991) modified this approach to consider the balance between the two energy dissipating systems in the lower compartment of the vessel because the energy dissipated by the agitator in the upper compartment of a large vessel is not related to gas dispersion.
- (iii) Feijen *et al.* (1987) claimed that flooding could be avoided if:

$$F_s/ND^3 < 0.3 N^2 D/g \quad (9.12)$$

where F_s is the volumetric air flow rate at the pressure conditions of the lower stirrer ($\text{m}^3 \text{second}^{-1}$), N is the stirrer speed (second^{-1}), D is the stirrer diameter (m), g is the gravitational acceleration (m second^{-2}).

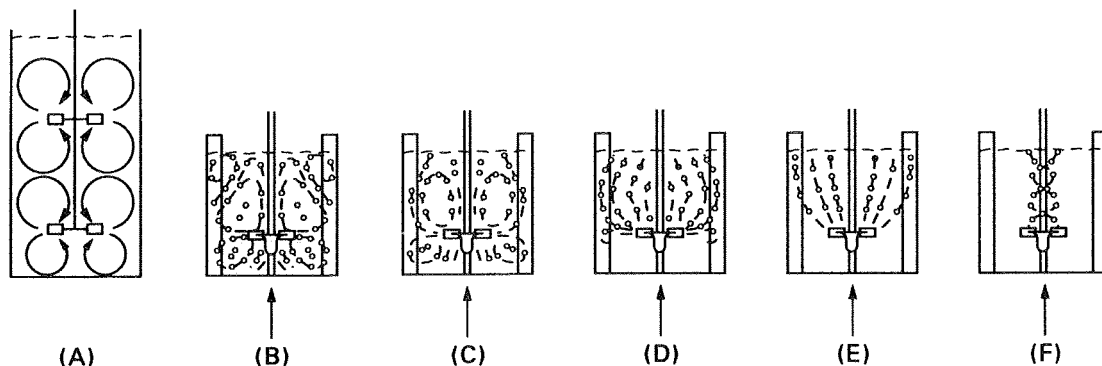


FIG. 9.13. The effect of air flow rate on the flow pattern in stirred vessels (After Nienow *et al.*, 1977).

A. Non-aerated;

B to F, increasing air flow rates.

NON-MECHANICALLY AGITATED REACTORS

Bubble columns and air-lift reactors are not mechanically agitated and, therefore, rely on the passage of air to both mix and aerate.

(i) *Bubble columns*

The flow pattern of bubbles through a bubble column reactor is dependent on the gas superficial velocity (cm second^{-1}). At gas velocities of below $1\text{--}4 \text{ cm second}^{-1}$ the bubbles will rise uniformly through the medium (Van't Riet and Tramper, 1991) and the only mixing will be that created in the bubble wake. This type of flow is referred to as homogeneous. At higher gas velocities bubbles are produced unevenly at the base of the vessel and bubbles coalesce resulting in local differences in fluid density. The differences in fluid density create circulatory currents and flow under these conditions is described as heterogeneous as shown in Fig. 9.14.

Flooding in a bubble column is the situation when the air flow is such that it blows the medium out of the vessel. This requires superficial gas velocities approaching 1 m second^{-1} which are not attainable on commercial scales (Van't Riet and Tramper, 1991).

The volumetric mass transfer coefficient ($K_L a$) in a bubble column is essentially dependent on the superficial gas velocity. Heijnen and Van't Riet (1984) reviewed the subject and demonstrated that the precise mathematical relationship between $K_L a$ and superficial gas velocity is dependent on the coalescent properties of the medium, the type of flow and the bubble size. Unfortunately these characteristics are rarely known for a commercial process which makes the ap-

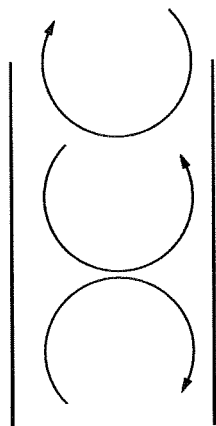


FIG. 9.14. Schematic representation of a heterogeneous flow regime in a bubble column.

plication of these equations problematical. However, Van't Riet and Tramper (1991) claimed that the relationship derived for non-coalescing, non-viscous, large bubbles (6 mm diameter) will give a reasonably accurate estimation for most non-viscous situations:

$$K_L a = 0.32 (V_s^c)^{0.7} \quad (9.13)$$

where V_s^c is the superficial air velocity corrected for local pressure.

However, viscosity has an overwhelming influence on $K_L a$ in a bubble column which Deckwer *et al.* (1982) expressed as:

$$K_L a = c\pi^{-0.84} \quad (9.14)$$

where π is the liquid dynamic viscosity (N s m^{-2}).

The practical implication of this equation is that bubble columns cannot be used with highly viscous fluids. Van't Riet and Tramper (1991) suggested that the upper viscosity limit for a bubble column was $100 \times 10^{-3} \text{ N s m}^{-2}$ at which point the $K_L a$ would have decreased 50 fold compared with a reactor batched with water.

(ii) *Air-lift reactors*

The structure of air-lift reactors is discussed in Chapter 7. The difference between a bubble column and an air-lift reactor is that liquid circulation is achieved in the air-lift in addition to that caused by the bubble flow. The reactor consists of a vertical loop of two connected compartments, the riser and downcomer. Air is introduced into the base of the riser and escapes at the top. The degassed liquid is more dense than the gassed liquid in the riser and flows down the downcomer. Thus, a circulatory pattern is established in the vessel — gassed liquid going up in the riser and degassed liquid coming down the downcomer.

For a given air-lift reactor and medium $K_L a$ varies linearly with superficial air velocity on a log-log scale over the normal range of velocities (Chen, 1990). However, it should be remembered that the circulation in an air-lift results in the bubbles being in contact with the liquid for a shorter time than in a corresponding bubble column. Thus, the $K_L a$ obtained in an air-lift will be less than that obtained in a bubble column at the same superficial air velocity, i.e. less than $0.32 (V_s^c)^{0.7}$. The advantage of the air-lift lies in the circulation achieved, but this is at the cost of a lower $K_L a$ value.

As for a bubble column flooding will not occur within the normal operating superficial air velocities and should not be a problem on a large scale.

The effect of the degree of agitation on $K_L a$

The degree of agitation has been demonstrated to have a profound effect on the oxygen-transfer efficiency of an agitated fermenter. Banks (1977) claimed that agitation assisted oxygen transfer in the following ways:

- (i) Agitation increases the area available for oxygen transfer by dispersing the air in the culture fluid in the form of small bubbles.
- (ii) Agitation delays the escape of air bubbles from the liquid.
- (iii) Agitation prevents coalescence of air bubbles.
- (iv) Agitation decreases the thickness of the liquid film at the gas-liquid interface by creating turbulence in the culture fluid.

The degree of agitation may be measured by the amount of power consumed in stirring the vessel contents. The power consumption may be assessed by using a dynamometer, by using strain gauges attached to the agitator shaft and by measuring the electrical power consumption of the agitator motor (see Chapter 8). The assessment of electrical consumption is suitable only for use with large-scale vessels.

THE RELATIONSHIP BETWEEN $K_L a$ AND POWER CONSUMPTION

A large number of empirical relationships have been developed between $K_L a$, power consumption and superficial air velocity which take the form of:

$$K_L a = k(P_g/V)^x V_s^y$$

where P_g is the power absorption in an aerated system

V is the liquid volume in the vessel

V_s is the superficial air velocity

k, x

and y are empirical factors specific to the system under investigation.

Cooper *et al.* (1944) measured the $K_L a$ s of a number of agitated and aerated vessels (up to a volume of 66 dm³) containing one impeller, using the sulphite oxidation technique, and derived the following expression:

$$K_L a = k(P_g/V)^{0.95} V_s^{0.67}. \quad (9.15)$$

Thus, it may be seen from equation (9.15) that the $K_L a$ value was claimed to be almost directly proportional to the gassed power consumption per unit volume. However, Bartholomew (1960) demonstrated that the relationship depended on the size of the vessel and the

exponent on the term P_g/V varied with scale as follows:

Scale	Value of exponent on P_g/V
Laboratory	0.95
Pilot plant	0.67
Production plant	0.5

Bartholomew's vessels contained more than one impeller, whereas those of Cooper *et al.* contained only one. It is probable that the upper impellers would consume more power relative to their contribution to oxygen transfer than would the lowest impeller, thus affecting the value of the exponent term. Thus, it is important to appreciate that such relationships are scale-dependent when using them in scale-up calculations.

Many workers have produced similar correlations and these have been reviewed by Van't Riet (1983) and Winkler (1990). Van't Riet (1983) summarized the various correlations for coalescing air-water dispersion systems as falling within 20–40% of:

$$k = 0.026,$$

$$x = 0.4,$$

$$y = 0.5.$$

The common feature of these relationships is that the values of x and y are less than unity. Winkler (1990) pointed out that this means that increasing power input or air flow becomes progressively less efficient as the inputs rise. Thus, high oxygen-transfer rates are achieved at considerable expense.

From this discussion it is evident that the $K_L a$ of an aerated, agitated vessel is affected significantly by the consumption of power during stirring and, hence, the degree of agitation. Although it is not possible to derive a relationship between $K_L a$ and power consumption which is applicable to all situations it is possible to derive a relationship between the two which is operable within certain limits and should be a useful guide in practical design problems. If it is accepted that such relationships between power consumption and $K_L a$ are of some practical significance, it is of considerable importance to relate power consumption to operating variables which may affect it. Quantitative relationships between power consumption and operating variables may be useful in:

- (i) Estimating the amount of power that an agita-

tion system will consume under certain circumstances, which could assist in fermenter design.

- (ii) In providing similar degrees of power consumption (and, hence, agitation and, therefore, K_L as) in vessels of different size.

THE RELATIONSHIP BETWEEN POWER CONSUMPTION AND OPERATING VARIABLES

Rushton *et al.* (1950) investigated the relationship between power consumption and operating variables in baffled, agitated vessels using the technique of dimensional analysis. They demonstrated that power absorption during agitation of non-gassed Newtonian liquids could be represented by a dimensionless group termed the power number, defined by the expression:

$$N_p = P / (\rho N^3 D^5) \quad (9.16)$$

where N_p is the power number,

P is the external power from the agitator,

ρ is the liquid density,

N is the impeller rotational speed,

D is the impeller diameter.

Thus, the power number is the ratio of external force exerted (P) to the inertial force imparted ($\rho N^3 D^5$) to the liquid. The motion of liquids in an agitated vessel may be described by another dimensionless number known as the Reynolds number which is a ratio of inertial to viscous forces:

$$N_{Re} = (\rho D^2 N) / \mu \quad (9.17)$$

where N_{Re} is the Reynolds number and

μ is the liquid viscosity.

Yet another dimensionless number, termed the Froude number, relates inertial force to gravitational force and is given the term:

$$N_{Fr} = (\rho N D^2) / g \quad (9.18)$$

where N_{Fr} is the Froude number and

g is the gravitational force.

Rushton *et al.* (1950) demonstrated that the power number was related to the Reynolds and Froude numbers by the general expression:

$$N_p = c (N_{Re})^x (N_{Fr})^y \quad (9.19)$$

where c is a constant dependent on vessel geometry

but independent of vessel size,

x and y are exponents.

Examples of the values of c , x and y are considered later.

However, in a fully baffled agitated vessel the effect of gravity is minimal so that the relationship between the power number and the other dimensionless num-

bers becomes:

$$N_p = c (N_{Re})^x \quad (9.20)$$

Therefore substituting from equations (9.16) and (9.17):

$$P / (\rho N^3 D^5) = c (\rho D^2 N / \mu)^x \quad (9.21)$$

Values for P at various values of N , D , μ and ρ may be determined experimentally and the Reynolds and power numbers for each experimental situation may then be calculated. A plot of the logarithm of the power number against the logarithm of the Reynolds number yields a graph termed the power curve. A typical power curve for a baffled vessel agitated by a flat-blade turbine is illustrated in Fig. 9.15 and such a curve would apply to geometrically similar vessels regardless of size.

From Fig. 9.15 it may be seen that a power curve is divisible into three clearly defined zones depicting different types of fluid flow:

- (i) The laminar or viscous flow zone where the logarithm of the power number decreases linearly with an increase in the logarithm of the Reynolds number. The slope of the graph is equal to x , the exponent in equation (9.21) and is obviously equal to -1 . The power absorbed in this region is a function of the viscosity of the liquid and the Reynolds number is less than 10.
- (ii) The transient or transition zone, where there is no consistent relationship between the power and Reynolds numbers. The value of x (that is, the slope of the plot) is variable and the value of the Reynolds number is between 10 and 10^4 .
- (iii) The turbulent flow zone, where the power number is a constant, independent of the

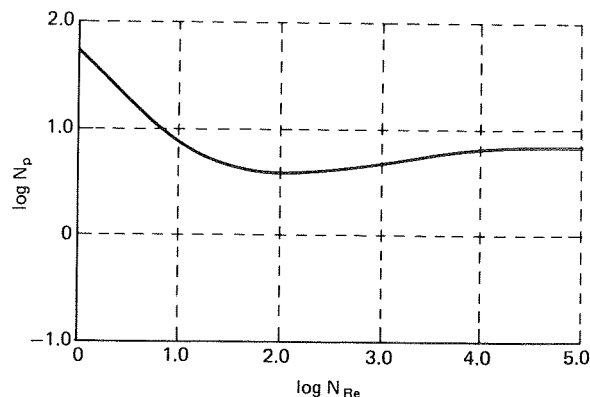


FIG. 9.15. A typical power curve for a baffled vessel agitated by a flat-blade turbine.

Reynolds number so that the value of x is zero and the value of the Reynolds number is in excess of 10^4 .

If the values of the exponent, x are substituted into equation (9.21) for the zones of viscous and turbulent flow, then the following terms are given:

$$\text{For viscous flow } P = c\mu N^2 D^3. \quad (9.22)$$

$$\text{For turbulent flow } P = c\rho N^3 D^5. \quad (9.23)$$

From these equations it may be seen that power consumption is dependent only on the viscosity of the liquid in the region of viscous flow and that increased speed of agitation, or an increase in the impeller diameter, results in a proportionally greater increase in power transmission to a liquid in turbulent flow than to one in viscous flow. Conditions of viscous flow are rare in fermentation processes, the majority of fermentations exhibiting flow characteristics in either the turbulent or transition zones. If turbulent flow is demonstrated to occur in a fermentation then equation (9.23) may be used to predict its power requirements and to predict the operating conditions of different sized vessels to achieve the same agitation conditions, as outlined by Banks (1979). Power consumption on the small scale may be represented as:

$$P_{sm} = c\rho N_{sm}^3 D_{sm}^5 \quad (9.24)$$

and on the large scale as:

$$P_L = c\rho N_L^3 D_L^5$$

where the subscripts sm and L refer to the small and large scales respectively. Maintaining the same power input per unit volume:

$$P_{sm}/P_L = V_{sm}/V_L = (c\rho N_{sm}^3 D_{sm}^5)/(c\rho N_L^3 D_L^5) \quad (9.25)$$

where V is the volume.

Assuming the vessels to be geometrically similar then c will be the same regardless of scale and as the same broth would be employed ρ would remain the same for both systems

$$V_{sm}/V_L = (N_{sm}^3 D_{sm}^5)/(N_L^3 D_L^5) \quad (9.26)$$

For geometrically similar vessels

$$D_{sm}/D_L = (V_{sm}/V_L)^{1/3}$$

Therefore, substituting for D_{sm}/D_L in (9.26)

$$N_L = N_{sm}(V_{sm}/V_L)^{2/9}. \quad (9.27)$$

If transient flow conditions occur in a fermentation then it is necessary to construct a complete power curve for such predictions and this is discussed later in the chapter.

The work of Rushton *et al.* (1950) was carried out using ungassed liquids whereas the vast majority of fermentations are aerated. It is widely accepted that aeration of a liquid decreases the power consumption during agitation because an aerated liquid, containing suspended air bubbles, is less dense than an unaerated one and large gas-filled cavities generated behind the agitator blades decrease the hydrodynamic resistance of the blades. A number of workers have produced correlations of gassed power consumption, ungassed power consumption and operating variables, that of Michel and Miller (1962) being widely used:

$$P_g = k(P^2 ND^3/Q^{0.56})^{0.45}$$

where Q is the volumetric air flow rate.

However, more recent correlations have been elucidated which are applicable over a wider range of operating conditions than that of Michel and Miller. Hughmark (1980) produced the following correlation from 248 sets of published data:

$$P_g/P = 0.1(Q/NV)^{-0.25}(N^2 D^4/gWV^{0.67})^{-0.2}$$

where Q is the volumetric air flow rate,
 g is the acceleration due to gravity,
 and W is the impeller blade width.
 Using dimensional analysis:

$$P_g/P = 0.0312 \cdot \text{Fr}^{-1.6} \cdot \text{Re}^{0.064} N_a^{-0.38} \cdot (T/D)^{0.8}$$

where N_a is the aeration number and equals Q/ND and T is the vessel diameter.

Provided it is remembered that these expressions are not particularly accurate they may be used to predict power consumption in gassed systems where turbulent flow is known to be operating. However, it should be remembered that in non-mycelial fermentations the greatest power demands often occur during agitation when the system is not gassed, that is during the sterilization of the medium *in situ* or if the air supply were to fail. Thus, in designing the system care must be taken to ensure that the agitator motor is sufficiently powerful to agitate the ungassed system and for fixed speed motors the operating speed should be specified with respect to the ungassed power draw (Gbewonyo *et al.*, 1986).

From the foregoing account it may be seen that reasonable techniques exist to relate operating vari-

ables to power consumption and, hence, to the degree of agitation which may be shown to have a proportional effect on $K_L a$. However, these techniques apply to Newtonian fluids and are not directly applicable to the study of non-Newtonian systems. Non-Newtonian fluids do not have constant viscosities, which creates difficulties in utilizing relationships which rely on being able to determine the fluid viscosity. These difficulties may be avoided if the agitation system is capable of maintaining turbulent-flow conditions during the fermentation, because under such conditions power consumption is independent of the Reynolds number and, hence, of viscosity. However the high viscosities of the majority of mycelial fermentation broths make fully turbulent flow conditions impossible, or extremely difficult, to achieve. Such fermentations tend to exhibit transient zone flow conditions which necessitate the construction of complete power curves to correlate power consumption with operating variables. The fact that the viscosity of a non-Newtonian liquid is affected by shear rate means that the viscosity of a non-Newtonian fermentation broth will not be uniform throughout the fermenter because the shear rate will be higher near the agitator than elsewhere in the vessel. Thus, the determination of the impeller Reynolds number is made difficult by not knowing the viscosity of the fermentation broth. Metzner and Otto (1957) proposed a solution to this paradox by introducing the concept of average shear rate ($\dot{\gamma}$) related to the agitator shaft speed in the vessel, by the equation:

$$\dot{\gamma} = kN \quad (9.28)$$

where k is a proportionality constant.

Metzner and Otto determined the value of the proportionality constant to be 13 for pseudoplastic fluids in conventional, baffled reactors agitated by single, flat-blade turbines. Several groups of workers have determined values of k under a wide range of operating variables; the values range from approximately 10 to 13. Metzner *et al.* (1961) suggested that a compromise value of 11 could be used for calculation purposes, with relatively little loss of accuracy, which would obviate the necessity to determine k for each circumstance. Therefore, provided that the rheological properties of a fermentation broth are known, an apparent average viscosity of the fluid may be calculated using the average shear rate which would enable the calculation of the impeller Reynolds number for each value of the impeller rotational speed, thus enabling a power curve to be constructed. Such a power curve may be used to predict the power requirements of a fermentation and to scale up a fermentation on the basis of

power consumption per unit volume. Metzner and Otto's approach has not been widely applied but there are some recent examples of its adoption. Nienow and Elson (1988) suggested that a repetition of their work would be very valuable using the more sophisticated instrumentation now available. An example of the use of the technique is considered in a later section considering the operation of viscous polysaccharide fermentations.

The effect of medium and culture rheology on $K_L a$

As can be seen from the previous section, the rheology of a fermentation broth has a marked influence on the relationship between $K_L a$ and the degree of agitation. The objective of this section is to discuss the effects of medium and culture rheology on oxygen transfer during a fermentation. A fermentation broth consists of the liquid medium in which the organism grows, the microbial biomass and any product which is secreted by the organism. Thus, the rheology of the broth is affected by the composition of the original medium and its modification by the growing culture, the concentration and morphology of the biomass and the concentration and rheological properties of the microbial products. Therefore, it should be apparent that fermentation broths vary widely in their rheological properties and significant changes in broth rheology may occur during a fermentation.

MEDIUM RHEOLOGY

Fermentation media frequently contain starch as a carbon source which may render the medium non-Newtonian and relatively viscous. However, as the organism grows it will degrade the starch and thus modify the rheology of the medium and reduce its viscosity. Such a situation was described by Tuffile and Pinho (1970) in their study of the growth of *Streptomyces aureofaciens* on a starch-containing medium. Before inoculation, the medium displayed Bingham plastic characteristics with a well-defined yield stress and an apparent viscosity of approximately 18 pseudopoise; after 22 hours the organism's activity had decreased the medium viscosity to a few pseudopoise and modified its behaviour to that of a Newtonian liquid; from 22 hours onwards the apparent viscosity of the broth gradually increased, due to the development of the mycelium, up to a maximum of approximately 90 pseudopoise and the rheology of the broth became increasingly pseudoplastic in nature. Thus, this example suggests that the rheological prob-

lems presented by the medium are minor compared with those presented by a high mycelial biomass, especially when it is considered that the total oxygen demand is relatively low in the early stages of a fermentation. However, it is worth remembering that in non-viscous unicellular fermentations the highest power draw will occur when the medium is sterilized *in situ* when the vessel is not being aerated and this will correspond with the time when a starch-based medium is at its most viscous.

Berkman-Dik *et al.* (1992) observed that the medium composition could affect the rheological properties of mycelium suspensions by affecting the interactions between the hyphae.

THE EFFECT OF MICROBIAL BIOMASS ON $K_L a$

(i) Agitator design for non-Newtonian fermentations

The biomass concentration and its morphological form in a fermentation has been shown to have a profound effect on oxygen transfer. Most bacterial and yeast fermentations tend to give rise to relatively non-viscous Newtonian broths in which conditions of turbulent flow may be achieved. Such fermentations present relatively few oxygen-transfer problems. However, the highly viscous non-Newtonian broths of fungal and streptomycete fermentations present major difficulties

in oxygen provision, the productivities of many such fermentations being limited by oxygen availability. Banks (1977) stressed the difference in the pattern of oxygen uptake between unicellular and mycelial fermentations as illustrated in Fig. 9.16. In both unicellular and mycelial fermentations the pattern of total oxygen uptake is very similar during the exponential growth phase, up to the point of oxygen limitation. However, during oxygen limitation, when arithmetic growth occurs, the oxygen uptake rate remains constant in a unicellular system whereas it decreases in a mycelial one. Banks claimed that the only possible explanation for such a decrease is the increasing viscosity of the culture caused by the increasing mycelial concentration.

Several groups of workers have demonstrated the detrimental effect of the presence of mycelium on oxygen transfer. Figure 9.17 represents some of the data of Deindoerfer and Gaden (1955) illustrating the effect of *Penicillium chrysogenum* mycelium on $K_L a$. Buckland *et al.* (1988), using different agitator systems, reported that the $K_L a$ decreased approximately in proportion with the square root of the broth viscosity, i.e:

$$K_L a \propto 1/\sqrt{\text{viscosity}}.$$

Steel and Maxon (1966) investigated the problem of oxygen provision to mycelial clumps in the *Streptomyces*

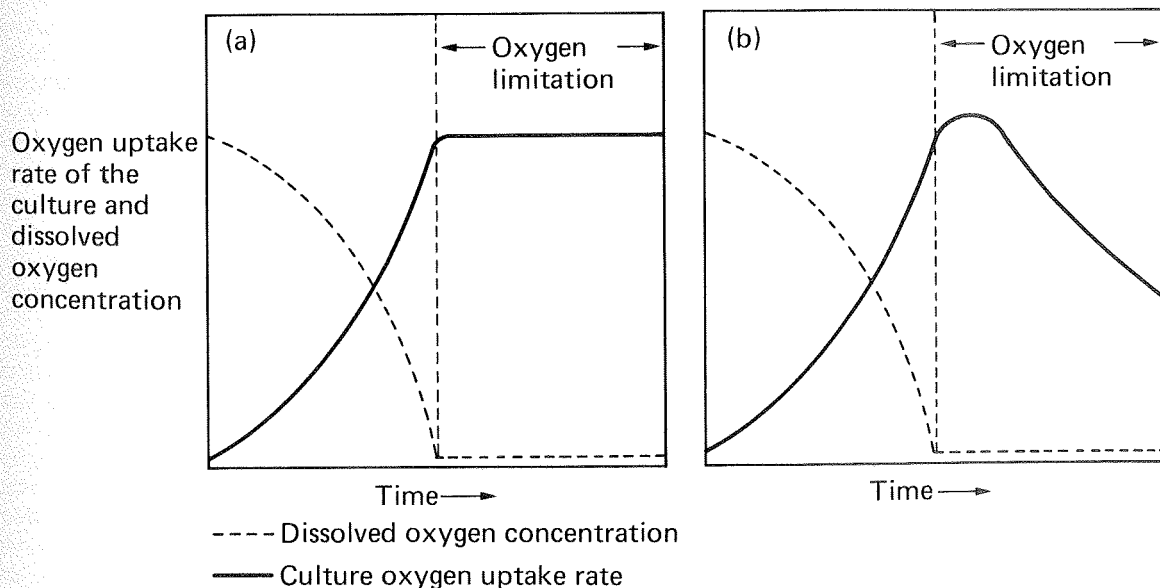


FIG. 9.16. The effect of oxygen limitation on the culture oxygen uptake rate: (a) A typical bacterial fermentation. (b) A typical fungal fermentation (Banks, 1977).

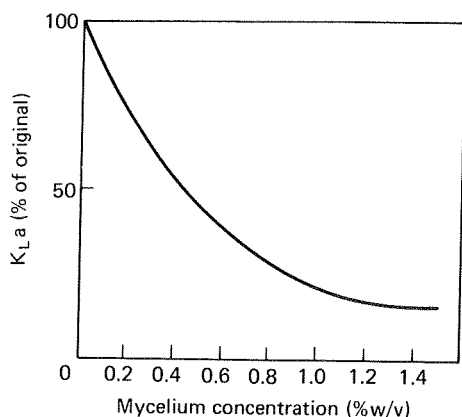


FIG. 9.17. The effect of *Penicillium chrysogenum* mycelium on $K_L a$ in a stirred fermenter (Deindoerfer and Gaden, 1955).

niveus novobiocin fermentation and demonstrated that high dissolved oxygen levels (60–80%) occurred in oxygen-limited cultures. It was concluded that, although oxygen was being transferred into solution, the dissolved gas was not reaching a large proportion of the biomass. Thus, as well as $K_L a$ being affected adversely by a high viscosity broth, efficient mixing also becomes extremely important in these systems. These workers also demonstrated that, at constant power input, small impellers were superior to large impellers in transferring oxygen from the gas phase to the microbial cells. Wang and Fewkes (1977) confirmed Steel and Maxon's work by demonstrating that the critical dissolved oxygen concentration (C_{crit}) for *S. niveus* in a fermentation varied depending on the degree of agitation and the size of the impeller. Remember that C_{crit} is the dissolved oxygen concentration below which oxygen uptake is limited, i.e. it is a physiological characteristic of the organism. It was concluded that the limiting factor was the diffusion of oxygen to the cell surface through a dense mycelial mass. At higher agitation rates biomass within clumps would be receiving oxygen and would thus contribute to the measured respiration rate whereas at low agitation rates such mycelium would be oxygen limited, i.e. the heterogeneity of the system increased at low agitation rates. Wang and Fewkes examined their results in terms of the impeller's ability to produce turbulent shear stress (oxygen transfer into solution) and pumping power (mixing). Turbulent shear stress is proportional to $N^2 D^2$ and impeller pumping power is proportional to ND^3 (where N is the impeller rotational speed and D is the impeller diame-

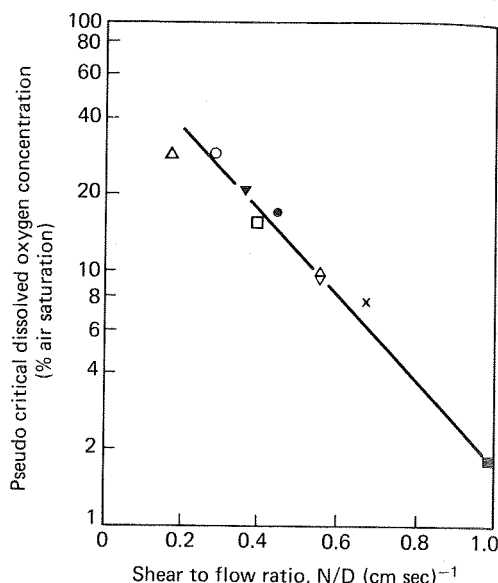


FIG. 9.18. The effect of shear to flow ratio on the observed critical oxygen concentration of *S. niveus* (Wang and Fewkes, 1977).

ter). Thus, the ratio of impeller turbulent shear stress to impeller pumping is proportional to:

$$N^2 D^2 / ND^3 \text{ or } N/D \text{ (cm sec)}^{-1}.$$

It was demonstrated that the observed critical dissolved-oxygen concentration decreased exponentially as the shear stress to pumping ratio increased, over the range 0.2 to 1.0 (cm sec^{-1}), as shown in Fig. 9.18. Thus, an increase in the ratio of impeller shear stress to impeller pumping decreases the transport resistance of oxygen to the cell surface resulting in a lower dissolved oxygen concentration maintaining a higher respiration rate. Wang and Fewkes' analysis quantifies Steel and Maxon's observation that smaller impellers gave better oxygen transfer to the cells of *S. niveus*, in that the smaller impeller would have a larger shear stress to impeller pumping power ratio.

Wang and Fewkes' correlations are particularly relevant when it is considered that many mycelial broths are pseudoplastic. The viscosity of a pseudoplastic broth will decrease with increasing shear stress so that viscosity increases with increasing distance from the agitator. Air introduced into the fermenter tends to rise through the vessel by the route of least resistance, that is, through the well-stirred, less viscous central zone. Thus, stagnant zones, receiving little oxygen, may occur in the vessel. Therefore, it is essential that the agitation regime employed creates the correct balance of turbu-

lence (and hence the transfer of oxygen into solution) and pumping power (mixing) to circulate the broth through the region of high shear.

The quantification of the problem of oxygen transfer and mixing is also considered by Van't Riet and Van Sonsberg (1992) in the context of the critical time for mass transfer. It is assumed that oxygen transfer into solution in a stirred, aerated reactor takes place only in the stirrer region. If one considers an aliquot of aerated broth leaving the agitator zone, it will be circulated through the vessel and eventually return to the agitator. The dissolved oxygen imparted to the broth should sustain the respiration of the organisms in that aliquot during the circulation. The time it takes for the oxygen in the aliquot to be exhausted will be:

$$t_{cro} = C_L(ag)/OUR$$

where t_{cro} is the time for oxygen to be exhausted,
 $C_L(ag)$ is the dissolved oxygen concentration in the zone of the agitator,
 OUR is the oxygen uptake rate.

If the circulation time for the vessel exceeds t_{cro} then oxygen starvation will occur in the aliquot before it returns to the agitator. To prevent this occurring the dissolved oxygen concentration at the agitator should be high, but this would reduce the driving force of oxygen into solution and the oxygen transfer rate would decrease. The alternative approach is to achieve the balance of mass transfer and pumping power (broth circulation) already discussed.

As discussed in Chapter 7 the most widely used fermenter agitator is a disc turbine (Rushton turbine). Van't Riet (1979) and Chapman *et al.* (1983) demonstrated that for non-viscous broths the $K_L a$ is dependent only on the power dissipated in the vessel and is independent of impeller type (at least those impellers included in the study). However, it is obvious from the foregoing discussion that impeller type is particularly relevant for viscous, non-Newtonian fermentations and this realization has resulted in the development of a range of agitators which address the dual problems of oxygen transfer and mixing in viscous fermentations.

Legrys and Solomons (1977) approached the problem of combining adequate pumping power and mass transfer in mycelial fermentations by using two impellers, a bottom-mounted disc turbine and a top-mounted curled-blade (hydrofoil) impeller. The bottom turbine produced a high degree of turbulence and radial mixing while the top-mounted impeller produced axial mixing with a high flow velocity, resulting in the circulation of one tank volume in 20–30 seconds. Thus, the mycelium was re-circulated through the oxygenation zone of the

vessel before it became oxygen limited. Cooke *et al.* (1988) extended Legrys and Solomon's approach using a combination of radial flow and axial flow agitators in a 60-dm³ fermenter intended for non-Newtonian fermentations. The radial flow agitator was an ICI Gasfoil which is similar to the Scaba SRGT illustrated in Fig. 9.19, being a disc turbine with concave blades. However, in this case the combination was not successful due to minimal fluid movement at the vessel walls which would have created significant cooling problems.

Gbewonyo *et al.* (1986) evaluated the performance of a hydrofoil impeller, the Prochem Maxflo (Fig. 9.19) in the avermectin fermentation employing *Streptomyces*

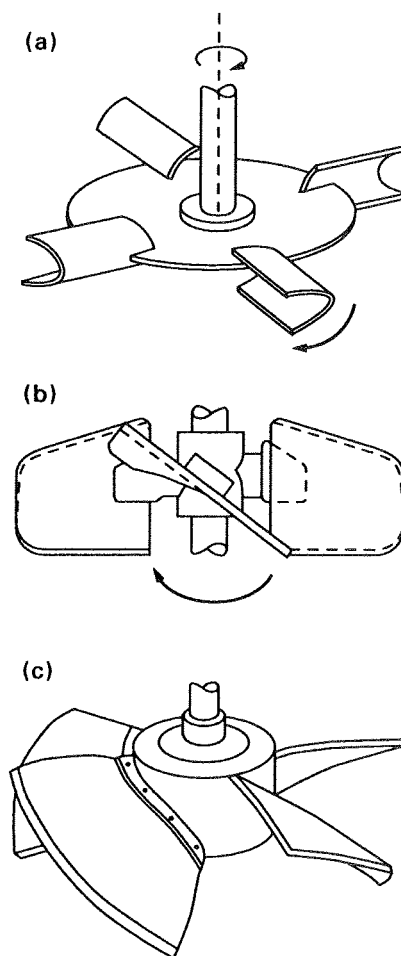


FIG. 9.19. Agitators used in filamentous fermentations:
 (a) Scaba agitator;
 (b) Lightning' A315;
 (c) Prochem Maxflo;
 (Nienow, 1990).

avermilis in a 600-dm³ working volume vessel. The avermectin process is challenging because the broth is extremely viscous, the fermentation requires fairly high oxygen-transfer rates and the situation is complicated by the shear sensitivity of the mycelium. The results of this investigation may be summarized as follows:

- (i) The impeller pumped the broth axially, that is from the top to the bottom of the fermenter, which is very different from the Rushton turbine which pumped radially, outwards from the agitator.
- (ii) The Prochem agitator supported a significantly higher oxygen uptake rate than did the Rushton turbine.
- (iii) The power number of the Prochem was 1.1 compared with 6.5 for the Rushton turbine. The equation (9.16) for power number was given previously as:

$$N_p = P / (\rho N^3 D^5)$$

where N_p is the power number,
 P is the external power from the agitator,
 ρ is the liquid density,
 N is the impeller rotational speed,
 D is the impeller diameter.

Thus, a low power number indicates a low power draw and, hence, the Prochem agitator drew significantly less power than did the Rushton turbine, making the former far more economical to operate. This observation is strengthened by Nienow's work on a similar hydrofoil impeller, the 'Lightnin' A315, which gave a power number of 0.75 compared with 5.2 for a Rushton turbine.

- (iv) The relationships between $K_L a$ and power consumption per unit volume at a viscosity of 700 cp were as follows:

$$\begin{aligned} \text{Rushton } K_L a &= 51(P/V)^{0.58}, \\ \text{Prochem } K_L a &= 129(P/V)^{0.59}. \end{aligned}$$

These figures reinforce the previous point, demonstrating that the power requirement for the Prochem agitator is approximately 50% of that for the Rushton.

- (v) Raising the power of the Prochem had a greater effect on oxygen transfer at high viscosity than it did at low viscosity. This points to the key role that bulk mixing plays in a viscous fermentation and suggests that it is at least as impor-

tant as bubble breakup (at which the Prochem is mediocre).

- (vi) Unlike a Rushton turbine, the Prochem agitator did not generate high shear forces, which is advantageous for a shear sensitive organism.
- (vii) The avermectin yields were slightly better in the Prochem fermenter, but these were achieved with approximately 40% less power consumption.

The same group (Buckland *et al.*, 1988, 1989) performed similar experiments on viscous fungal fermentations in 800-dm³ and 19-m³ vessels and came to the same basic conclusions that bulk mixing is extremely important in viscous fermentations and that an axial flow hydrofoil impeller results in lower power costs. Data generated from non-Newtonian polysaccharide fermentations using hydrofoil impellers are considered in a subsequent section of this chapter.

(ii) *The manipulation of mycelial morphology*

The previous section considered engineering solutions to the problem of oxygen transfer in mycelial fermentations. However, this is not the only approach to improve oxygen transfer in such processes; it is possible to modify the morphology of the process organism. As discussed in Chapter 6, the biomass of mycelial organisms grown in submerged culture may vary from the filamentous type, in which the hyphae form a homogeneous suspension dispersed through the medium, to the 'pellet' type consisting of compact, discrete masses of hyphae. The filamentous form tends to give rise to a highly viscous, non-Newtonian broth whereas the pellet form tends to produce an essentially Newtonian system with a much lower viscosity making oxygen transfer much easier. Buckland (1993) reported that the $K_L a$ attained in the lovastatin *Aspergillus terreus* fermentation was 20 h⁻¹ with a filamentous culture and 80 h⁻¹ with a pelleted one at the same power input. Not all pelleted cultures are Newtonian: Metz *et al.* (1979) demonstrated that pellet suspensions could be non-Newtonian but confirmed that they did give rise to low viscosity broths. Also, it should be appreciated that the terms 'filamentous' and 'pelleted' each describe a range of morphology and the form of filamentous or pelleted growth may be affected by both the genetic makeup of the organism and the environment. Thus, the morphological form of a mycelial organism in submerged culture has a major effect on the broth rheology and may, therefore, be expected to influence aeration efficiency.

Carilli *et al.*'s work (1961) provides a good example

of both the effect of different filamentous form on process performance and the behaviour of filamentous and pelleted cultures in 3000-dm³ fermenters. Two strains of *P. chrysogenum* were employed, one which grew as short, highly branched hyphae and the other as long, relatively unbranched hyphae. The short, branched hyphae gave rise to a relatively low viscosity broth in which the oxygen transfer rate was approximately twice that achieved with the more viscous broth of the unbranched form. By manipulating the cultural conditions of *A. niger*, Carilli *et al.* were able to produce the fungus in either filamentous or pellet form and demonstrated that the pellet form gave rise to a broth exhibiting half the viscosity of the filamentous broth. Also, oxygen limitation occurred far earlier in the fermentation when the organism grew in the filamentous form.

Although the pellet type of growth tends to produce a low viscosity Newtonian broth in which turbulent flow conditions may be achieved, it may also give rise to problems of oxygen availability if the pellets become too large. A large pellet may be so compact that its centre may be unaffected by the turbulent forces occurring in the bulk of the fermentation broth so that the passage of oxygen within the pellet is dependent on simple diffusion; this may result in the centre of the pellet being oxygen limited. Thus, to maintain the intra-pellet oxygen concentration at an adequate level it would be necessary to maintain a high dissolved oxygen concentration to ensure an effective diffusion gradient. A similar situation was described by Steel and Maxon (1966) and Wang and Fewkes (1977). Kobayashi *et al.* (1973) demonstrated this phenomenon in pellets of *A. niger* where large pellets required a higher dissolved oxygen concentration to maintain the same specific oxygen-uptake rate as smaller pellets. If oxygen limitation does occur within a pellet then only its outer layer would contribute to its growth and the centre may autolyse. The diffusion of oxygen into the centre of a pellet will be influenced by the size of the pellet, and thus it is important to control pellet size.

Schugerl *et al.* (1988) monitored the dissolved oxygen concentration within pellets of *P. chrysogenum* and demonstrated that, provided they were smaller than 400 μm in diameter, the oxygen concentration in the centre of the pellet was not limiting. Similarly, Buckland (1993) reported that pellets of *Aspergillus terreus* in the lovastatin fermentation had to be smaller than 180 μm in diameter to avoid oxygen limitation in the centre of the pellet. It should be appreciated that the pellet sizes recommended by both groups are very small and it is possible to obtain fungal pellets which

are at least 1 cm in diameter. Pellet size may be influenced by the inoculum, the medium and the cultural conditions. As discussed in Chapter 6, pellet size is reduced at high spore inoculum concentrations, but it is unlikely that this alone would produce pellets of less than 400 μm in diameter. Schugerl *et al.* (1988) controlled the pellet size of the inoculum by physical means by either incorporating glass beads in inoculum shake flasks or using high agitator speeds in seed fermenters. Metz and Kossen (1977) also claimed that, once pellets are formed, strong agitation tends to give rise to smaller, more compact pellets. Buckland (1993) reported that the conditions of the lovastatin fermentation are carefully controlled to maintain the optimum pellet size but the cultural conditions used to achieve this end were not revealed.

Righelato (1979) discussed the effects of mycelium morphology on culture rheology and oxygen transfer and came to the conclusion that the most desirable way for a mycelium to grow in submerged culture is in the form of short, hyphal fragments which would produce a broth less susceptible to diffusion limitation than a pelleted one, and less viscous than one containing long filaments. However, attempts to encourage the formation of the desirable short hyphal fragment morphology (as compared with the long filaments) by increasing the shear stress on the mycelium has met with only limited success. Even if a less viscous broth is obtained the damage done to the mycelium may well be counter-productive. Dion *et al.* (1954) showed that the morphology of *P. chrysogenum* was influenced by the degree of agitation in that short, branched mycelium was produced at high agitation rates compared with long hyphae produced at low agitation rates. Lilly *et al.* (1992) extended this observation at 10 dm³ and 100 dm³ scales and related the mean main hyphal length and the penicillin specific production rate (q_p) to the term $P/D_i^3 t_c$, where P is the agitator power, D_i is the impeller diameter and t_c is the calculated circulation time. This term is a measure of the maximum shear stress due to agitator power dissipation and the frequency with which mycelia pass through the high shear region. Both mean hyphal length and q_p decreased with increasing $P/D_i^3 t_c$, implying that increased shear is disadvantageous at this scale. At 1000 dm³ it was not possible to introduce enough power into the fermenter to decrease q_p , which suggests that it is very difficult to disrupt the mycelium at this scale, thus confirming Van Suijdam and Metz's (1981) observation that an enormous amount of energy is required to reduce the hyphal length of *P. chrysogenum*. Righelato (1979) also claimed that it is unlikely that shear forces could ac-

count for the break up of mycelia and that autolysis and lysis of some hyphal compartments may be more important controlling factors, perhaps implying that the phenomenon may be more under genetic, rather than physical, control. This leads us on to strain improvement of morphologically favourable strains, as discussed in more detail in Chapter 3. However, Belmar-Beiny and Thomas (1990) demonstrated in 9-dm³ fermenters that increased stirrer speed did result in the production of shorter, less branched hyphal fragments of *Streptomyces clavuligerus* and clavulanic acid synthesis was unaffected. This suggests that this approach may be used to influence rheological properties in clavulanic acid fermentations.

Other cultural conditions which have been claimed to influence mycelial morphology include medium composition (see Chapter 5), growth rate, dissolved oxygen concentration, polymer additives and temperature. Kuenzi (1978) reported that the viscosity of a *Cephalosporium* broth was considerably reduced by growing the organism at 27° rather than 25°C. Olsvik and Kristiansen (1992) investigated the influence of specific growth rate and dissolved oxygen concentration on the viscosity of *Aspergillus niger* in continuous culture. *K*, the consistency index (indicative of apparent viscosity, see equation (9.9)) was measured over a range of conditions. At dissolved oxygen (DO) concentrations above 10% saturation, *K* increased with increasing dilution rate whereas at DO concentrations below 10% saturation, *K* decreased with increasing dilution rate. The effect of DO on *K* was particularly evident at low DO values and at low growth rates where a 2% change in the DO could give a 25% change in *K*. These observations may be particularly relevant in the late stages of batch or fed-batch processes where low growth rates, nutrient limitation, high biomass levels and low oxygen concentrations occur, all contributing to complex changes in morphology, viscosity and oxygen transfer rate.

Dispersed growth can be encouraged in certain organisms by incorporating polymeric compounds into the medium. Such anionic polymers include Junlon PW110 and Junlon 111 (cross-linked polyacrylic acids) and Carbopol-934 (carboxypolymethylene). It is claimed that these polymers modify the electrical charges on the spore surface and thus prevent the aggregation of spores into clumps, thus preventing the initiation of pellet formation. These agents have been used to increase the homogeneity of both fungal (Trinci, 1983) and streptomycete (Hobbs *et al.*, 1989) broths. Although these agents would not be practical to use on a large scale they may be useful in the early stages of an

inoculum development programme if a dispersed morphology is desirable.

Several workers have discussed the possible advantages of reducing the viscosity of a mycelial fermentation, in its later stages, by diluting the broth with either water or fresh medium. Sato (1961) increased the yield of a kanamycin fermentation, displaying Bingham plastic rheology, by 20% by diluting the broth 5% by volume with sterile water. Taguchi (1971) achieved a 50% reduction in the viscosity of an *Endomyces* broth by diluting 10% with water or fresh medium. A scheme has been put forward for the control of viscosity and dissolved oxygen concentration in a hypothetical fermentation. These workers proposed that, as the critical dissolved oxygen concentration is approached, a set volume of broth could be removed from the fermenter and replaced with fresh medium. The process could be repeated in a step-wise manner as the system became oxygen limited, which could be determined by dissolved oxygen concentration or viscosity measurements. Thus, by using such techniques the viscosity may be controlled and maintained below the level which may cause oxygen limitation. Kuenzi (1978) reported an instance where the very slow feeding of medium to a *Cephalosporium* culture resulted in the organism growing in the form of long filaments which produced a highly viscous culture which could not be adequately aerated. The design of fed-batch processes such that efficient control may be achieved over the process is discussed in a subsequent section of this chapter and in Chapter 2.

The production of *Fusarium gramineae* biomass for human food in the ICI-RHM mycoprotein (Quorn®) fermentation (see Chapter 1) presents a very different problem from those of most other fungal fermentations. It is essential that the organism grows as long hyphae so that the biomass can be processed into a textured food product. Long hyphae are susceptible to shear forces, so to maintain the morphological form of the organism an air-lift reactor is used, despite the fact that the viscous broth severely limits the attainable oxygen transfer rate. This limitation of the air-lift fermenter means that only a relatively low biomass concentration may be maintained in the vessel compared with that in a stirred system, but this is an acceptable penalty to pay for the correct morphological form.

THE EFFECT OF MICROBIAL PRODUCTS ON AERATION

Generally speaking, the product of a fermentation contributes relatively little to the viscosity of the cul-

ture broth. However, the exception is the production of bacterial polysaccharides, where the broths tend to be highly viscous (30,000 cp, Sutherland and Ellwood, 1979) and non-Newtonian. Charles (1978) demonstrated that the bacterial cells in a polysaccharide fermentation made a minimal contribution to the high culture viscosity which was due primarily to the polysaccharide product. Normally, microbial polysaccharides tend to behave as pseudoplastic fluids, although some have also been shown to exhibit a yield stress. The yield stress of a polysaccharide can make the fermentation particularly difficult because, beyond a certain distance from the impeller, the broth will be stagnant and productivity in these regions will be practically zero (Gallindo and Nienow, 1992). Thus, bacterial polysaccharide fermentations present problems of oxygen transfer and bulk mixing similar to those presented by mycelial fermentations. Thus, similar stirrer configurations to those discussed in the previous section have been used in polysaccharide fermentations. Gallindo and Nienow (1992) investigated the behaviour of a hydrofoil impeller, the Lightnin' A315, in a simulated xanthan fermentation. These workers adopted Metzner and Otto's approach to construct power curves. Better agitator performance was achieved when its pumping direction was upwards rather than downwards resulting in lower power loss on aeration and less torque fluctuations. It was concluded that such agitators may give improved mixing in a xanthan fermentation provided that the polysaccharide concentration is below 25 kg m^{-3} .

A novel solution to the problem was proposed by Oosterhuis and Koerts (1987). These workers designed an air-lift loop reactor incorporating a pump to circulate the highly viscous broth. The system was operated on a 4-m^3 scale and proved to be much more efficient than a stirred tank reactor.

The effect of foam and antifoams on oxygen transfer

The high degree of aeration and agitation required in a fermentation frequently gives rise to the undesirable phenomenon of foam formation. In extreme circumstances the foam may overflow from the fermenter via the air outlet or sample line resulting in the loss of medium and product, as well as increasing the risk of contamination. The presence of foam may also have an adverse effect on the oxygen-transfer rate. Hall *et al.* (1973) pointed out that Waldhof and vortex-type fermenters (see Chapter 7) were particularly affected due to the bubbles becoming entrapped in the continuously

recirculating foam, resulting in high bubble residence times and, therefore, oxygen-depleted bubbles. The presence of foam in a conventional agitated, baffled fermenter may also increase the residence time of bubbles and therefore result in their being depleted of oxygen. Furthermore, the presence of foam in the region of the impeller may prevent adequate mixing of the fermentation broth. Thus, it is desirable to break down a foam before it causes any process difficulties and, as discussed in Chapter 7, this may be achieved by the use of mechanical foam breakers or chemical antifoams. However, mechanical foam control consumes considerable energy and is not completely reliable so that chemical antifoams are preferred (Van't Riet and Van Sonsberg, 1992).

All antifoams are surfactants and may, themselves, be expected to have some effect on oxygen transfer. The predominant effect observed by most workers is that antifoams tend to decrease the oxygen-transfer rate, as discussed by Aiba *et al.* (1973) and Hall *et al.* (1973). Antifoams cause the collapse of bubbles in foam but they may favour the coalescence of bubbles within the liquid phase, resulting in larger bubbles with reduced surface area to volume ratios and hence a reduced rate of oxygen transfer (Van't Riet and Van Sonsberg, 1992). Thus, a balance must be struck between the necessity for foam control and the deleterious effects of the controlling agent. Foam formation has a particular influence on the liquid height in the fermenter at which it is practical to operate. If inadequate space is provided above the liquid level for foam control, then copious amounts of antifoam must be used to prevent loss of broth from the vessel. Van't Riet and Van Sonsberg (1992) observed that, above a critical liquid height, the $K_L a$ value decreases dramatically due to the excessive use of antifoams. Thus, it may be more productive to operate a vessel at a lower working volume.

Methods for foam control are considered in Chapter 8 and antifoams are discussed in Chapter 4.

THE BALANCE BETWEEN OXYGEN SUPPLY AND DEMAND

Both the demand for oxygen by a micro-organism and the supply to the organism by the fermenter have been considered in this chapter. This section attempts to bring these two aspects together and considers how processes may be designed such that the oxygen uptake rate of the culture does not exceed the oxygen transfer rate of the fermenter.

The volumetric oxygen uptake rate of a culture is described by the term, $Q_{O_2}x$, where Q_{O_2} is the specific oxygen uptake rate (mmoles O_2 g⁻¹ biomass h⁻¹) and x is biomass concentration (g dm⁻³). Thus, the units of $Q_{O_2}x$ are mmoles oxygen dm⁻³ h⁻¹.

The volumetric oxygen transfer rate (also measured as mmoles O_2 dm⁻³ h⁻¹) of a fermenter is given by equation (9.1), i.e.:

$$dC_L/dt = K_L a(C^* - C_L).$$

It will also be recalled that the dissolved oxygen concentration during the fermentation should not fall below the critical dissolved oxygen concentration (C_{crit}) or the dissolved oxygen concentration which gives optimum product formation. Thus, it is necessary that the oxygen-transfer rate of the fermenter matches the oxygen uptake rate of the culture whilst maintaining the dissolved oxygen above a particular concentration. A fermenter will have a maximum $K_L a$ dictated by the operating conditions of the fermentation and thus, to balance supply and demand it must be the demand that is adjusted to match the supply. This may be achieved by:

- (i) Controlling biomass concentration.
- (ii) Controlling the specific oxygen uptake rate.
- (iii) A combination of (i) and (ii).

Controlling biomass concentration

Mavituna and Sinclair (1985a) developed a method to predict the highest biomass concentration (termed the critical biomass or x_{crit}) which can be maintained under fully aerobic conditions in a fermenter of known $K_L a$. Thus, x_{crit} is the biomass concentration which gives a volumetric uptake rate ($Q_{O_2}x_{crit}$) equal to the maximum transfer rate of the fermenter, i.e. $K_L a(C^* - C_{crit})$. If C_{crit} is defined as the dissolved oxygen concentration when:

$$Q_{O_2} = 0.99Q_{O_2} \max$$

then the volumetric oxygen uptake rate when the dissolved oxygen concentration is C_{crit} will be:

$$0.99Q_{O_2} \max \cdot x_{crit}.$$

If the oxygen transfer rate were equal to the uptake rate when the dissolved oxygen concentration equals C_{crit} , then:

$$K_L a(C^* - C_{crit}) = 0.99Q_{O_2} \max \cdot x_{crit}. \quad (9.29)$$

Equation (9.29) may be used to calculate x_{crit} for a fermenter with a particular $K_L a$ value:

$$x_{crit} = K_L a(C^* - C_{crit})/0.99 Q_{O_2} \max \quad (9.30)$$

Equation (9.30) may also be modified to calculate the biomass concentration which may be maintained at any fixed dissolved oxygen concentration above C_{crit} :

$$x = K_L a(C^* - C_L)/Q_{O_2} \max.$$

Mavituna and Sinclair presented this model graphically as shown in Fig. 9.20. The upper graph represents the relationship between the dissolved oxygen concentration and the volumetric oxygen transfer rate achievable in three fermenters (plots 1, 2 and 3 represent fermenters of increasing $K_L a$ values) whilst the lower graph represents the relationship between biomass and the volumetric oxygen uptake rate of the culture. The x axes of both graphs are drawn to the same scale. A construction is drawn on the upper graph linking C_{crit} to the oxygen-transfer rates attainable in each of the three fermenters. This construction is extended to the

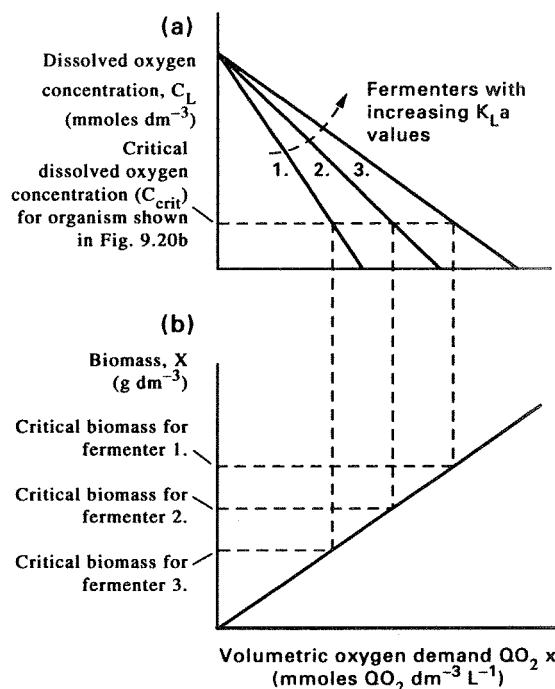


FIG. 9.20. (a) The relationship between dissolved oxygen concentration and the oxygen transfer rate attainable in 3 fermenters with increasing $K_L a$ values. (b) The relationship between biomass concentration and oxygen uptake rate of a process organism. The same scales are used for dC_L/dt and $Q_{O_2}x$ allowing x_{crit} to be determined (Mavituna and Sinclair, 1985).

lower graph indicating the oxygen uptake rates equal to the transfer rates attainable at C_{crit} . Finally, from the lower graph the biomass concentrations (x_{crit}) which would give rise to the uptake rates equal to the transfer rates may be determined. Again, this figure may be used to predict the maximum biomass concentration which may be maintained at any dissolved oxygen concentration above C_{crit} .

It should be appreciated that these authors intended this model to be used only as a method for preliminary design (Mavituna and Sinclair, 1985b). Thus, x_{crit} is interpreted as a target which cannot be exceeded and, in practice, oxygen limitation will probably occur below this value. The mechanism for limiting the biomass concentration will be the concentration of the limiting substrate in the medium which, for batch culture, may be determined from the equation:

$$S_R = x_{crit}/Y$$

where S_R is the initial limiting substrate concentration and Y is the yield factor and it is assumed that the limiting substrate is exhausted on entry into the stationary phase.

The technique may also be applied to continuous and fed-batch culture but it must be appreciated that Q_{O_2} is affected by specific growth rate and the relevant Q_{O_2} value for the growth rate employed would have to be utilized in the calculations. The method should be very useful for the initial design of unicellular bacterial or yeast fermentations where biomass has no effect on $K_L a$. However, in viscous fermentations the biomass concentration influences the $K_L a$ considerably, as discussed in a previous section. Thus, the $K_L a$ will decline with increasing biomass concentration which makes the application of the technique more problematical.

Controlling the specific oxygen uptake rate

Specific oxygen-uptake rate is directly proportional to specific growth rate so that, as μ increases, so does Q_{O_2} . Thus, Q_{O_2} may be controlled by the dilution rate in continuous culture. Although very few commercial fermentations are operated in continuous culture, fed-batch culture is widely used in industrial fermentations and provides an excellent tool for the control of oxygen demand. The kinetics and applications of fed-batch culture are discussed in Chapter 2. The most common way in which the technique is applied to control oxygen demand is to link the nutrient addition system to a feed-back control loop using a dissolved oxygen electrode as the sensing element (see Chapter 8). If the dissolved oxygen concentration declines below the set

point then the feed rate is reduced and when the dissolved oxygen concentration rises above the set point the feed rate may be increased. A pH electrode may also be used as a sensing unit in a fed-batch control loop for the control of oxygen demand — oxygen limitation being detected by the development of acidic conditions. These techniques are particularly important in the growth-stage of a secondary metabolite mycelial fermentation prior to product production when the highest growth rate commensurate with the oxygen transfer rate of the fermenter is required. A full discussion of the operation of fed-batch systems is given in Chapter 2.

SCALE-UP AND SCALE-DOWN

Scale-up means increasing the scale of a fermentation, for example from the laboratory scale to the pilot plant scale or from the pilot plant scale to the production scale. Increase in scale means an increase in volume and the problems of process scale-up are due to the different ways in which process parameters are affected by the size of the unit. It is the task of the fermentation technologist to increase the scale of a fermentation without a decrease in yield or, if a yield reduction occurs, to identify the factor which gives rise to the decrease and to rectify it. The major factors involved in scale-up are:

- (i) *Inoculum development.* An increase in scale may mean that extra stages have to be incorporated into the inoculum development programme. This aspect is considered in Chapter 6.
- (ii) *Sterilization.* Sterilization is a scale dependent factor because the number of contaminating micro-organisms in a fermenter must be reduced to the same absolute number regardless of scale. Thus, when the scale of a process is increased the sterilization regime must be adjusted accordingly, which may result in a change in the quality of the medium after sterilization. This aspect is considered in detail in Chapter 5.
- (iii) *Environmental parameters.* The increase in scale may result in a changed environment for the organism. These environmental parameters may be summarized as follows:

- (a) nutrient availability,
- (b) pH,

- (c) temperature,
- (d) dissolved oxygen concentration,
- (e) shear conditions,
- (f) dissolved carbon dioxide concentration,
- (g) foam production.

All the above parameters are affected by agitation and aeration, either in terms of bulk mixing or the provision of oxygen. Points a, b, c and e are related to bulk mixing whilst d, e, f and g are related to air flow and oxygen transfer. Thus, agitation and aeration tends to dominate the scale-up literature. However, it should always be remembered that inoculum development and sterilization difficulties may be the reason for a decrease in yield when a process is scaled up and that achieving the correct aeration/agitation regime is not the only problem to be addressed.

Scale-up of aeration/agitation regimes in stirred tank reactors

From the list of environmental parameters affected by aeration and agitation it will be appreciated that it is extremely unlikely that the conditions of the small-scale fermentation will be replicated precisely on the large scale. Thus, the most important criteria for a particular fermentation must be established and the scale-up based on reproducing those characteristics. The problem of aeration/agitation scale-up has been extremely well illustrated by Fox (1978) in his description of the 'scale-up window'. The scale-up window represents the boundaries imposed by the environmental parameters and cost on the aeration/agitation regime and is shown in Fig. 9.21. Suitable conditions of mixing and oxygen transfer can be obtained with a range of aeration/agitation combinations. The two axes of Fig. 9.21 are agitation and aeration and the zone within the hexagon represents suitable aeration/agitation regimes. The boundary of the hexagon is defined by the limits of oxygen supply, carbon dioxide accumulation, shear damage to the cells, cost, foam formation and bulk mixing. For example, the agitation rate must fall between a minimum and maximum value — mixing is inadequate below the minimum level and shear damage to the cells is too great above the maximum value. The limits for aeration are determined at the minimum end by oxygen limitation and carbon dioxide accumulation and at the maximum end by foam formation. The shape of the window will depend on the fermentation — for exam-

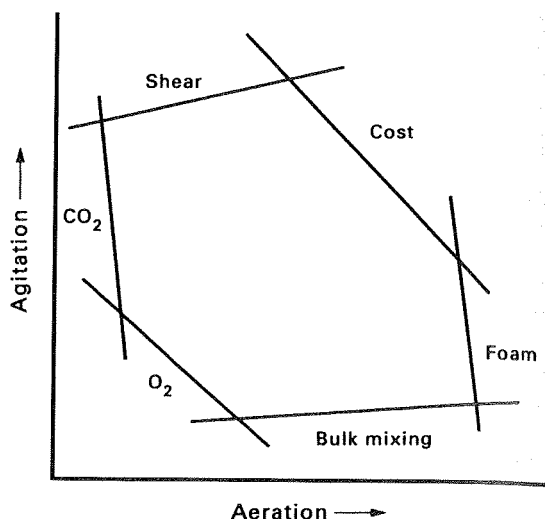


FIG. 9.21. The 'scale-up' window defining the operating boundaries for aeration and agitation in the scale-up of a fermentation. After Fox (1978) reproduced from Lilly (1983).

ple, the supply of oxygen would be irrelevant in an anaerobic fermentation, whereas the limitation due to shear would be of major importance in the scale-up of animal cell fermentations.

The solution of the scale-up problem is three-fold:

- (i) The identification of the principal environmental domain affected by aeration and agitation in the fermentation, e.g. oxygen concentration, shear, bulk mixing.
- (ii) The identification of a process variable (or variables) which affects the identified environmental domain.
- (iii) The calculation of the value of the process variable to be used on the large scale which will result in the replication of the same environmental conditions on both scales.

The process variables which affect mixing and mass transfer are summarized in Table 9.6 (Oldshue, 1985; Scragg, 1991). Thus, if dissolved oxygen concentration is perceived as the over-riding environmental condition then power consumption per unit volume and volumetric air flow rate per unit volume should be maintained constant on scale-up. However, as a result, the other parameters will not be the same in the larger scale and, therefore, neither will the environmental factors which they influence. This phenomenon is well illustrated by Oldshue's example summarized in Table 9.7 where a 125 fold increase in scale is represented. If power

TABLE 9.6. The effect of process variables on mass transfer or mixing characteristics

Process variable	Mass transfer or mixing characteristic affected
Power consumption per unit volume	Oxygen-transfer rate
Volumetric air flow rate	Oxygen-transfer rate
Impeller tip speed	Shear rate
Pumping rate	Mixing time
Reynolds number (see previous section)	Heat transfer

consumption per unit volume is kept constant then impeller tip speed (i.e. shear) increases and flow min^{-1} (i.e. mixing) decreases. If mixing is kept constant, an enormous (and totally uneconomic) increase in power is required and shear increases 5 fold. If impeller tip speed (shear) is kept constant then power consumption (hence, $K_L a$) and mixing decrease. This analysis indicates that it is economically impossible to maintain the same degree of mixing on scale-up and, therefore, a decrease in yield may be due to mixing anomalies.

The most important environmental domains affected by aeration and agitation for the majority of fermentations are oxygen concentration and shear. Thus, the most widely used scale-up criteria are the maintenance of a constant $K_L a$ or constant shear conditions. Constant shear may be achieved by scaling up on the basis of constant impeller tip speed. Constant $K_L a$ may be achieved on the basis of constant power consumption per unit volume and constant volumetric air-flow rate. The operating variable dictating constant power consumption in geometrically similar vessels is the agitator speed. The agitator speed on the large scale is then calculated from the correlations between $K_L a$ and power consumption and between power consumption and operating variables. An example of this approach is given in the previous section describing the effects of operating variables on power consumption.

Hubbard (1987) and Hubbard *et al.* (1988) summarized the procedure for scaling up both Newtonian and non-Newtonian fermentations and proposed two methods to determine the large scale conditions:

Method 1

- (i) Determine the volumetric air flow rate (Q) on the large scale based on maintaining Q/V constant (V = working volume of the fermenter).
- (ii) Calculate the agitator speed that will give the same $K_L a$ on the large scale; this is achieved using the correlations between power consumption and N and between $K_L a$ and power consumption.

Method 2

- (i) Calculate the agitator speed keeping the impeller tip speed constant, πND_i .
- (ii) Calculate Q from power correlations and $K_L a$ correlations.

The accuracy of these scale-up techniques is only as good as the power and $K_L a$ correlations, so it is worth expending some considerable time to test the validity of potential correlations for the fermentation in question.

TABLE 9.7. The effect of the choice of scale-up criteria on operating conditions in the scaled-up vessel. Based on scale-up from 80 dm^3 to 10^4 dm^3 (Based on Oldshue, 1985)

Criterion used in scale-up	Effect on the operating conditions on the large scale (Large scale value/Small scale value)			
	P	P/V	Flow $\text{min}^{-1} \text{ vol}^{-1}$	ND_i
P/V	125.0	1.0	0.34	1.7
Flow $\text{min}^{-1} \text{ vol}^{-1}$	3125.0	25.0	1.0	5.0
ND_i (Impeller tip speed)	25.0	0.2	0.2	1.0
Reynolds number	0.2	0.0016	0.04	0.2

The scale-up of air-lift reactors

Bubble columns and air-lift vessels tend to be scaled-up on the basis of geometric similarity and constant gas velocity (Scragg, 1991). Under these conditions the K_La and shear rate in the two scales will be similar. The major difference will be the height of the vessels resulting in increased pressure at the base of the larger vessel. This would result in higher oxygen and carbon dioxide solubility which would give a higher K_La but might result in carbon dioxide inhibition. The other problem in the scale-up of air-lift systems is that the organism is exposed to extremes of oxygen levels in the riser and downcomer and the effects of these conditions should be investigated on the laboratory scale.

Scale-down methods

Scale-down is the situation where laboratory- or pilot-scale experiments are conducted under conditions which mimic the industrial-scale conditions. This approach is important in both the development of a new product and the improvement of an existing full-scale fermentation. The procedure has been reviewed by Jem (1989). Frequently, conditions achievable on a laboratory scale are impractical on an industrial scale, which means that if inappropriate conditions have been used in the laboratory unrealistic yield objectives may be set for the scaled-up process. The aspects to consider in the design of laboratory- or pilot-plant experiments in the context of scale-down may be summarized as follows:

- (i) *Medium design.* Media relevant to the industrial situation should be used in development experiments.
- (ii) *Medium sterilization.* If the medium is to be batch sterilized on the large scale its exposure time at a high temperature will be much greater than that experienced in the laboratory or pilot plant. Thus, the sterilization times on the smaller scales should be increased to mimic the industrial situation. Alternatively, medium sterilized in the production fermenter may be used in the laboratory and pilot plant. This highlights the advantage of continuous sterilization where little loss of medium quality occurs. Furthermore, the same continuous sterilizer may be used for both full-scale and pilot scale vessels.

- (iii) *Inoculation procedures.* Due to a range of circumstances, it may not always be possible to inoculate every production fermentation with inoculum in optimum condition. The scale-down approach can be used to predict the consequences of such events by mimicking these situations in the laboratory, for example by storing inoculum or using inocula of different ages.
- (iv) *Number of generations.* An industrial scale fermentation requires a greater number of generations than does a laboratory one; this may place more severe stability criteria on the process strain than may have been appreciated on the small scale. The industrial situation may be modelled in the laboratory by using serial sub-culture to ensure that the strain is sufficiently stable. This approach is particularly pertinent in the development of recombinant fermentations.
- (v) *Mixing.* As indicated in the previous section it is almost inevitable that the degree of mixing will decrease with an increase in scale. Thus, it is possible to model inadequate mixing in the laboratory by subjecting the organism to pulse medium feeds or fluctuating process conditions such as oxygen concentration, pH and temperature. Such scaled-down experiments then allow predictions to be made about the suitability of new strains for industrial exploitation.
- (vi) *Oxygen transfer rate.* Far higher oxygen transfer rates can be achieved in laboratory fermenters than in industrial-scale ones. Thus, unrealistic demands may be made of a fermentation plant if the development work has been done at very high oxygen-transfer rates. Therefore, the laboratory and pilot fermenters should reflect the oxygen transfer rates achievable in the full-scale fermenters.

The adoption of these simple approaches to small scale experimentation can prevent many scale-up problems before they even occur!

REFERENCES

- AIBA, S., HUMPHREY, A. E. and MILLIS, N. (1973) *Biochemical Engineering*. Academic Press, London.
- BANKS, G. T. (1977) Aeration of moulds and streptomycete culture fluids. *Topics in Enzyme and Fermentation Biotechnology*, Vol. 1, pp. 72–110 (Editor Wiseman, A.). Ellis Horwood, Chichester.

- BANKS, G. T. (1979) Scale-up of fermentation processes. *Topics in Enzyme and Fermentation Biotechnology*, Vol. 3, pp. 170–267 (Editor Wiseman, A.). Ellis Horwood, Chichester.
- BARTHOLOMEW, W. H. (1960) Scale-up of submerged fermentations. *Adv. App. Micro.* **2**, 289–300.
- BARTHOLOMEW, W. H., KARROW, E. O., SFAT, M. R. and WILHELM, R. H. (1950) Oxygen transfer and agitation in submerged fermentations. Mass transfer of oxygen in submerged fermentations of *Streptomyces griseus*. *Ind. Eng. Chem.* **42** (9), 1801–1809.
- BELL, G. H. and GALLO, M. (1971) Effect of impurities on oxygen transfer. *Process Biochem.* **6** (4), 33–35.
- BELMAR-BEINY, M. T. and THOMAS, C. R. (1990) Morphology and clavulanic acid production of *Streptomyces clavuligerus*: effect of stirred speed in batch fermentation. *Biotech. Bioeng.* **37** (5), 456–462.
- BERKMAN-DIK, T., OZILGEN, M. and BOZOGLU, T. F. (1992) Salt, EDTA, and pH effects on rheological behavior of mold suspensions. *Enzyme Microb. Technol.* **14**, 944–948.
- BISECKER, B. O. (1972) *Begasen von Flüssigkeiten mit Ruhrem.* VDI, Forschungsheft.
- BUCKLAND, B. C. (1993) Mevinolin production. Paper presented at the Soc. Gen. Microbiol. 124th Meeting, University of Kent, Canterbury, January 1993.
- BUCKLAND, B. C., GBEWONYO, K., JAIN, D., GLAZOMITSKY, K., HUNT, G. and DREW, S. W. (1988) Oxygen transfer efficiency of hydrofoil impellers in both 800L and 1900L fermenters. In *Proceedings of the 2nd. International Conference on Bioreactor Fluid Dynamics*, pp. 1–16 (Editor, King, R.). Elsevier, London.
- BUCKLAND, B. C., GBEWONYO, K., HALLADA, T., KAPLAN, L. and MASUREKAR, P. (1989) In *Novel Microbial Products in Medicine and Agriculture*, pp. 161–169 (Editor, Demain, A. L.). Elsevier, Amsterdam.
- CARILLI, A., CHAIN, E. B., GAULANDI, G. and MORISI, G. (1961) Aeration studies III. Continuous measurements of dissolved oxygen in fermentation broths — Rheology and mass transfer. *Biotech. Bioeng.* **18**, 745–790.
- CASSON, N. (1959) *Rheology of Disperse Systems*. Pergamon Press, Oxford.
- CHAPMAN, C. M., NIENOW, A. W., COOKE, M. and MIDDLETON, J. C. (1983) Particle–gas–liquid mixing in stirred vessels. Part 3. *Chem. Eng. Res. Des.* **61** (3), 167–181.
- CHARLES, M. (1978) Technical aspects of the rheological properties of microbial cultures. *Adv. Biochem. Eng.* **8**, 1–62.
- CHEN, N. Y. (1990) The design of airlift fermenters for use in biotechnology. *Biotechnol. Gen. Eng. Rev.* **8**, 379–396.
- COOKE, M., MIDDLETON, J. C. and BUSH, J. R. (1988) Mixing and mass transfer in filamentous fermentations. In *Proceedings of the 2nd International Conference on Bioreactor Fluid Dynamics*, pp. 37–64 (Editor, King, R.). Elsevier, London.
- COONEY, C. L. (1979) Conversion yields in penicillin production: Theory versus practice. *Process Biochem.* **14** (5), 31–33.
- COOPER, C. M., FERNSTROM, G. A. and MILLER, S. A. (1944) Performance of agitated gas–liquid contacters. *Ind. Eng. Chem.* **36**, 504–509.
- DARLINGTON, W. A. (1964) Aerobic hydrocarbon fermentation — A practical evaluation. *Biotech. Bioeng.* **6** (2), 241–242.
- DECKWER, W. D., NGUYEN-TIEN, K., SCHUMPE, A. and SERPEMEN, Y. (1982) Oxygen mass transfer into aerated CMC solutions in a bubble column. *Biotech. Bioeng.* **24**, 461–481.
- DEINDOERFER, F. H. and GADEN, E. L. (1955) Effects of liquid physical properties on oxygen transfer in penicillin fermentation. *Appl. Micro.* **3**, 253–257.
- DEINDOERFER, F. H. and WEST, J. M. (1960) Rheological examination of some fermentation broths. *J. Microbiol. Biochem. Technol. Eng.* **2**, 165–175.
- DION, W. M., CASILLI, A., SERMONTI, G. and CHAIN, E. B. (1954) The effect of mechanical agitation on the morphology of *Penicillium chrysogenum* Thom in stirred fermenters. *Rend. Ist Super Sanita* **17**, 187–205.
- FEIJEN, J., HEIJNEN, J. J. and VAN'T RIET, K. (1987) Gas hold-up and flooding in large-scale fermenters. In *Proceedings of Symposium on mixing and dispersion processes. Inst. Chem. Eng. and KIVI/NIRIA*. Delft Technical University.
- FOX, R. I. (1978) The applicability of published scale-up criteria to commercial fermentation processes. *Proc. 1st Eur. Cong. Biotechnol.*, Part 1, pp. 80–83.
- GALLINDO, E. and NIENOW, A. W. (1992) Mixing of highly viscous simulated fermentation broths with the Lightnin' A-315 impeller. *Biotechnol. Prog.* **8**, 233–239.
- GBEWONYO, K., DIMASI, D. and BUCKLAND, B. C. (1986) The use of hydrofoil impellers to improve oxygen transfer efficiency in viscous mycelial fermentations. In *International Conference on Bioreactor Fluid Dynamics*, pp. 281–299. BHRA, Cranfield, U.K.
- GOLDBERG, I., ROCK, J. S., BEN-BASSAT, A. and MATELES, R. I. (1976) Bacterial yields on methanol, methylamine, formaldehyde and formate. *Biotech. Bioeng.* **18**, 1657–1668.
- HALL, M. J., DICKINSON, S. D., PRITCHARD, R. and EVANS, J. I. (1973) Foams and foam control in fermentation processes. *Prog. Ind. Micro.* **12**, 171–234.
- HEIJNEN, J. J., RIET, K. W. and WOLTHUIS, A. J. (1980) Influence of very small bubbles on the dynamic $K_L a$ measurement in viscous gas–liquid systems. *Biotech. Bioeng.* **22**, 1945–1956.
- HEIJNEN, J. J. and VAN'T RIET, K. (1984) Mass transfer, mixing and heat transfer phenomena in low viscosity bubble column reactors. *Chem. Eng. J.* **28**, B21.
- HEINEKEN, F. G. (1970) Use of fast-response dissolved oxygen probes for oxygen transfer studies. *Biotech. Bioeng.* **12**, 145–154.
- HEINEKEN, F. G. (1971) Oxygen mass transfer and oxygen respiration rate measurements utilising fast response oxygen electrodes. *Biotech. Bioeng.* **13**, 599–618.
- HIROSE, Y. and SHIBAI, H. (1980) Effect of oxygen on amino acids fermentation. *Advances in Biotechnology*, Vol. 1, pp.

- 329–333 (Editors Moo-Young, M., Robinson, C. W. and Vezina, C.). Pergamon Press, Toronto.
- HOBBS, G., FRAZER, C. M., GARDNER, D. C. J., CULLUM, J. A. and OLIVER, S. G. (1989) Dispersed growth of *Streptomyces* in liquid culture. *Appl. Microbiol. Biotechnol.* **31** (3), 272–277.
- HUBBARD, D. W. (1987) Scale-up strategies for bioreactors containing non-Newtonian broths. *Ann. N.Y. Acad. Sci.*, **506**, 600–607.
- HUBBARD, D. W., HARRIS, L.R. and WIERENGA, M. K. (1988) Scale-up for polysaccharide fermentation. *Chem. Eng. Prog.* **84** (8), 55–61.
- HUGHMARK, G. A. (1980) Power requirements and interfacial area in gas–liquid turbine agitated systems. *Ind. Eng. Chem. Proc. Des. Dev.* **19**, 638–645.
- JEM, K. J. (1989) Scale-down techniques in fermentation. *Biopharmacology* **2**, 30–39.
- JOHNSON, M. J. (1964) Utilisation of hydrocarbons by micro-organisms. *Chem. Ind.* **36**, 1532–1537.
- KOBAYASHI, T., VAN DEDEM, G. and MOO-YOUNG, M. (1973) Oxygen transfer into mycelial pellets. *Biotech. Bioeng.* **15**, 27–45.
- KUENZI, M. T. (1978) Process design and control in antibiotic fermentation. In *Antibiotics and Other Secondary Metabolites, Biosynthesis and Production, FEMS Symp.* Vol. 5, pp. 39–56 (Editors Hutter, R., Leisinger, T., Neusch and Wehrli, W.). Academic Press, London.
- LEGRYS, G. A. and SOLOMONS, G. L. (1977) US patent application 23128.
- LILLY, M. D. (1983) Problems in process scale-up. In *Bioactive Microbial Products 2. Development and Production*, pp. 79–90 (Editors Nisbet, L. J. and Winstanley, D. J.). Academic Press, London.
- LILLY, M. D., ISON, A. and SHAMLOU, P. A. (1992) The influence of the physical environment in fermenters on antibiotic production by micro-organisms. In *Harnessing Biotechnology in the 21st Century* (Editors Ladisch, M. R. and Bose, A.). American Chemical Society, Washington, DC.
- MATELES, R. I. (1971) Calculation of the oxygen required for cell production. *Biotech. Bioeng.* **13** (4), 581–582.
- MATELES, R. I. (1979) The physiology of single cell protein (SCP) production. Soc. Gen. Microbiology Symposium, **29**, *Microbial Technology: Current State, Future Prospects*, pp. 29–52 (Editors, Bull. A. T., Ellwood, D. C. and Ratledge, C.). Cambridge University Press, Cambridge.
- MAVITUNA, F. and SINCLAIR, C. G. (1985a) A graphical method for the determination of critical biomass concentration for non-oxygen limited growth. *Biotechnol. Lett.* **7**, 69–74.
- MAVITUNA, F. and SINCLAIR, C. G. (1985b) Reply to a comment on 'A graphical method for the determination of critical biomass concentration for non-oxygen limited growth'. *Biotechnol. Lett.* **7**, 813–814.
- METZ, B. and KOSSEN, N. W. F. (1977) Pellet growth of moulds. *Biotech. Bioeng.* **19**, 781–799.
- METZ, B., KOSSEN, N. W. F. and VAN SUJDAM, J. C. (1979) Rheology of mould suspensions. *Adv. Biochem. Eng.* **11**, 103–156.
- METZNER, A. B. and OTTO, R. E. (1957) Agitation of non-Newtonian fluids. *A.I.Ch.E.J.* **3** (1), 3–10.
- METZNER, A. B., FEEHS, R. H., RAMOS, H. L., OTTO, R. E. and TOOTHILL, J. D. (1961) Agitation of viscous Newtonian and non-Newtonian fluids. *A.I.Ch.E.J.* **7**, 3–9.
- MICHELL, B. J. and MILLER, S. A. (1962) Power requirements of gas–liquid agitated systems. *A.I.Ch.E.J.* **8**, 262–266.
- NIENOW, A. W. and ELSON, T. P. (1988) Aspects of mixing in rheologically complex liquids. *Chem. Eng. Res. Des.*, **66**, 5–15.
- NIENOW, A. W., WISDOM, D. J. and MIDDLETON, J. C. (1977) The effect of scale and geometry on flooding, recirculation and power in gassed stirred vessels. Paper F1, *2nd. Eur. Conf. on Mixing*, March 1977, Cambridge.
- OLDSHUE, J. Y. (1985) Current trends in mixer scale-up techniques. In *Mixing of Liquids by Mechanical Agitation* pp. 309–341 (Editors Ulbrecht, J. and Patterson, G. K.). Gordon and Breach, New York.
- OLSVIK, E. S. and KRISTIANSEN, B. (1992) Influence of oxygen tension, biomass concentration and specific growth rate on the rheological properties of a filamentous fermentation broth. *Biotech. Bioeng.* **40**, 1293–1299.
- OOSTERHUIS, N. M. G. and KOERTS, K. (1987) Method and reactor vessel for the fermentation production of polysaccharides in particular, xanthan. *Eur. Pat. Appl.* EP 249,288.
- RIGHELATO, R. C. (1979) The kinetics of mycelial growth. In *Fungal Walls and Hyphal Growth*, pp. 1385–1402 (Editors Burnett, J. H. and Trinci, A. P. J.). Cambridge University Press, Cambridge.
- RIGHELATO, R. C., TRINCI, A. P. J., PIRT, S. J. and PEAT, A. (1968) Influence of maintenance energy and growth rate on the metabolic activity, morphology and conidiation of *Penicillium chrysogenum*. *J. Gen. Micro.* **50** (1), 394–412.
- RIVIERE, J. (1977) *Industrial Application of Microbiology* (translated and edited by Moss, M. O. and Smith, J. E.). Surrey University Press, Guildford.
- ROELS, J. A., VAN DENBERG, J. and VONKEN, R. M. (1974) The rheology of mycelial broths. *Biotech. Bioeng.* **16**, 181–208.
- RUSHTON, J. H., COSTICH, E. W. and EVERETT, H. J. (1950) Power characteristics of mixing impellers. *Chem. Eng. Prog.* **46**, 395–404.
- SATO, K. (1961) Rheological studies on some fermentation broths. (IV) Effect of dilution rate on rheological properties of fermentation broth. *J. Ferment. Technol.* **39**, 517–520.
- SCHUGERL, K., WITTLER, R. and LORENZ, T. (1988) The use of moulds in pellet form. *Trends Biotechnol.* **1** (4), 120–122.
- SCHULZE, K. L. and LIPE, R. S. (1964) Relationship between substrate concentration, growth rate and respiration rate of *Escherichia coli* in continuous culture. *Archiv. Mikrobiol.*, **48**, 1–20.
- SCRAGG, A. H. 1991 *Bioreactors in Biotechnology. A Practical Approach*. Ellis Horwood, Chichester.
- STEEL, R. and MAXON, W. D. (1966) Studies with a multiple-rod mixing impeller. *Biotech. Bioeng.* **8**, 109–116.

- SUTHERLAND, I. W. and ELLWOOD, D. C. (1979) Microbial exopolysaccharides — industrial polymers of current and future potential. In Soc. Gen. Microbiology Symposium, **29**, *Microbial Technology: Current State, Future Prospects*, pp. 107–150 (Editors, Bull, A. T., Ellwood, D. C. and Ratledge, C.). Cambridge University Press, Cambridge.
- TAGUCHI, H. (1971) The nature of fermentation fluids. *Adv. Biochem. Eng.* **1**, 1–30.
- TAGUCHI, H. and HUMPHREY, A. E. (1966) Dynamic measurement of the volumetric oxygen transfer coefficient in fermentation systems. *J. Ferm. Technol.* **44** (12), 881–889.
- TAGUCHI, H., IMANAKA, T., TERAMOTO, S., TAKATSU, M. and SATO, M. (1968) Scale-up of glucamylase fermentation by *Endomyces* sp. *J. Ferm. Technol.* **46** (10), 823–828.
- TRINCI, A. P. J. (1983) Effect of Junlon on the morphology of *Aspergillus niger* and its use in making turbidity measurements of microbial growth. *Trans. Br. Mycol. Soc.* **58**, 467–473.
- TUFFILE, C. M. and PINHO, F. (1970) Determination of oxygen transfer coefficients in viscous streptomycete fermentations. *Biotech. Bioeng.* **12**, 849–871.
- VAN SUJDAM, J. C. and METZ, B. (1981) Influence of engineering variables upon morphology of filamentous moulds. *Biotech. Bioeng.* **23**, 111–148.
- VAN'T RIET, K. (1979) Review of measuring methods and results in non-viscous gas-liquid mass transfer in stirred vessels. *Ind. Eng. Chem. Process Des. Dev.* **18** (3), 357–360.
- VAN'T RIET, K. (1983) Mass transfer in fermentation. *Trends Biotechnol.*, **1** (4), 113–116.
- VAN'T RIET, K. and TRAMPER, J. (1991) *Basic Bioreactor Design*. Marcel Dekker, New York.
- VAN'T RIET, K. and VAN SONSBERG, (1992) Foaming, mass transfer and mixing: Interrelations in large scale fermentations. In *Harnessing Biotechnology for the 21st Century*, pp. 189–192 (Editors Ladisch, M.R. and Bose, A.). American Chemical Society, Washington, DC.
- WANG, D. I. C. and FEWKES, R. C. J. (1977) Effect of operating and geometric parameters on the behaviour of non-Newtonian mycelial antibiotic fermentations. *Dev. Ind. Micro.*, **18**, 39–57.
- WANG, D. I. C., COONEY, C. L., DEMAINE, A. L., DUNNILL, P., HUMPHREY, A. E. and LILLY, M. D. (1979) *Fermentation and Enzyme Technology*. Wiley, New York.
- WERNAU, W. C. and WILKE, C. R. (1973) New method for evaluation of dissolved oxygen probe response for $K_L a$ determination. *Biotech. Bioeng.* **15**, 571–578.
- WESTERTERP, K. R., VAN DIERENDONCK, L. L. and DE KRAA, J. A. (1963) Interfacial areas in agitated gas liquid contactors. *Chem. Eng. Sci.* **18**, 157–169.
- WINKLER, M. A. (1990) Problems in fermenter design and operation. In *Chemical Engineering Problems in Biotechnology*, pp. 215–350 (Editor Winkler, M. A.). SCI/Elsevier, London.
- WISE, W. S. (1951) The measurement of the aeration of culture media. *J. Gen. Micro.* **5**, 167–177.
- WODZINSKI, R. S. and JOHNSON, M. J. (1968) Yield of bacterial cells from hydrocarbons. *Appl. Micro.* **16**, 1886–1891.
- ZHOU, W., HOLZHAUER-RIEGER, K., DORS, M. and SCHUGERL, K. (1992) Influence of dissolved oxygen concentration on the biosynthesis of cephalosporin C. *Enzyme Microb. Technol.* **14**, 848–854.

The Recovery and Purification of Fermentation Products

INTRODUCTION

The choice of recovery process is based on the following criteria:

THE EXTRACTION and purification of fermentation products may be difficult and costly. Ideally, one is trying to obtain a high-quality product as quickly as possible at an efficient recovery rate using minimum plant investment operated at minimal costs. Unfortunately, recovery costs of microbial products may vary from as low as 15% to as high as 70% of the total manufacturing costs (Aiba *et al.*, 1973; Swartz, 1979; Pace and Smith, 1981; Atkinson and Sainter, 1982; Datar, 1986). Obviously, the chosen process, and therefore its relative cost, will depend on the specific product. Atkinson and Mavituna (1991) indicate percentage of total costs being 15% for industrial ethanol, 20–30% for bulk penicillin G and up to 70% for enzymes. The high (and sometimes dominant) cost of downstream processing will affect the overall objective in some fermentations.

If a fermentation broth is analysed at the time of harvesting it will be discovered that the specific product may be present at a low concentration in an aqueous solution that contains intact micro-organisms, cell fragments, soluble and insoluble medium components and other metabolic products. The product may also be intracellular, heat labile and easily broken down by contaminating micro-organisms. All these factors tend to increase the difficulties of product recovery. To ensure good recovery or purification, speed of operation may be the overriding factor because of the labile nature of a product. The processing equipment must therefore be of the correct type and also the correct size to ensure that the harvested broth can be processed within a satisfactory time limit.

1. The intracellular or extracellular location of the product.
2. The concentration of the product in the fermentation broth.
3. The physical and chemical properties of the desired product (as an aid to selecting separation procedures).
4. The intended use of the product.
5. The minimal acceptable standard of purity.
6. The magnitude of bio-hazard of the product or broth.
7. The impurities in the fermenter broth.
8. The marketable price for the product.

The main objective of the first stage for the recovery of an extracellular product is the removal of large solid particles and microbial cells usually by centrifugation or filtration (Fig. 10.1). In the next stage, the broth is fractionated or extracted into major fractions using ultrafiltration, reverse osmosis, adsorption/ion-exchange/gel filtration or affinity chromatography, liquid-liquid extraction, two phase aqueous extraction or precipitation. Afterwards, the product-containing fraction is purified by fractional precipitation, further more precise chromatographic techniques and crystallization to obtain a product which is highly concentrated and essentially free from impurities. Other products are isolated using modifications of this flow-stream.

Attempts to simplify this outline extraction procedure for antibiotic recovery using 'whole broth' processing have met with limited success. The technique of

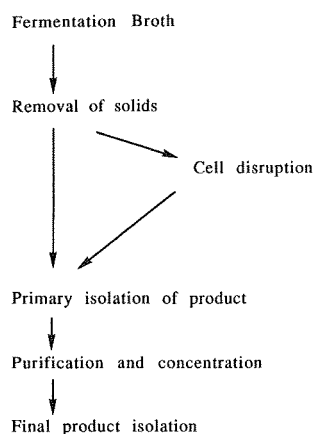


FIG. 10.1. Stages in the recovery of product from a harvested fermentation broth.

1. Harvest broth from fermenter
- ↓
2. Chill to 5-10°C
- ↓
3. Filter off *P. chrysogenum* mycelium using rotary vacuum filter
- ↓
4. Acidify filtrate to pH 2.0-2.5 with H₂SO₄
- ↓
5. Extract penicillin from aqueous filtrate into butyl acetate in a centrifugal counter-current extractor (treat/dispose aqueous phase)
- ↓
6. Extract penicillin from butyl acetate into aqueous buffer (pH 7.0) in a centrifugal counter-current extractor (recover and recycle butyl acetate)
- ↓
7. Acidify the aqueous fraction to pH 2.0-2.5 with H₂SO₄ and re-extract penicillin into butyl acetate as in stage 5
- ↓
8. Add potassium acetate to the organic extract in a crystallization tank to crystallize the penicillin as the potassium salt
- ↓
9. Recover crystals in a filter centrifuge (recover and recycle butyl acetate)
- ↓
10. Further processing of penicillin salt

FIG. 10.2. Recovery and partial purification of penicillin G.

'whole broth' processing involves initial removal of large particles, which is then followed by passage of the broth (including cells) through, for example, well mixed ion-exchange columns or counter-current liquid-liquid extraction units to extract the product directly. This topic will be discussed in more detail in a later section of this chapter.

It may be possible to modify the handling character-

istics of the broth so that it can be handled faster with simpler equipment making use of a number of techniques:

1. Selection of a micro-organism which does not produce pigments or undesirable metabolites.
2. Modification of the fermentation conditions to reduce the production of undesirable metabolites.
3. Precise timing of harvesting.
4. pH control after harvesting.
5. Temperature treatment after harvesting.
6. Addition of flocculating agents.
7. Use of enzymes to attack cell walls.

It must be remembered that the fermentation and product recovery are integral parts of an overall process. Because of the interactions between the two, neither stage should be developed independently, as this might result in problems and unnecessary expense. Darbyshire (1981) has considered this problem with reference to enzyme recovery. The parameters to consider included time of harvest, pigment production, ionic strength and culture medium constituents. Large volumes of supernatants containing extracellular enzymes need immediate processing while harvesting times and enzyme yields might not be predictable. This can make recovery programmes difficult to plan. Pigment production might make some recovery procedures difficult, when the pigment binds to the same resin as the enzyme. Changes in fermentation conditions may reduce pigment formation. Certain antifoams remain in the supernatant and affect ultrafiltration or ion-exchange resins used in recovery stages. Trials may be needed to find the most suitable antifoam (see also Chapter 4). The ionic strength of the production medium may be too high, resulting in the harvested supernatant needing dilution with demineralized water before it can be processed. Such a negative procedure should be avoided if possible by unified research and development programmes. Media formulation is dominated by production requirements, but the protein content of complex media should be critically examined in view of subsequent enzyme recovery. This view is also shared by Topiwala and Khosrovi (1978), when considering water recycle in biomass production. They stated that the interaction between the different unit operations in a recycle process made it imperative that commercial plant design and operation should be viewed in an integrated fashion.

Flow sheets for recovery of penicillin, cephamycin C,

1. Fermenter broth containing cephamycin C
- ↓
2. Adjust pH to 2.5
- ↓
3. Conventional filtration
- ↓
- Waste filter cake
- ↓
4. Cation exchange (sulfonic acid resin)
- ↓
5. Pyridine elution
- ↓
6. Concentration by evaporation (remove pyridine)
- ↓
7. Adjust pH to 5 - 7
- ↓
8. Anion exchange (tertiary amine)
- ↓
9. Alkanoic acid wash
- ↓
10. Pyridine or phosphate buffer solution

FIG. 10.3. Purification of cephamycin C: sequential ion exchange process (Omstead *et al.*, 1985).

citric acid and micrococcal nuclease are given in Figs 10.2, 10.3, 10.4 and 10.5, to illustrate the range of techniques used in microbiological recovery processes. A series of comprehensive flow sheets for alcohols, organic acids, antibiotics, carotenoids, polysaccharides, intra- and extra-cellular enzymes, single-cell proteins and vitamins have been produced by Atkinson and Mavituna (1991). Other reviews on separation and purification are available for penicillin (Swartz, 1979), amino acids (Samejima, 1972), enzymes (Aunstrup, 1979; Darbyshire, 1981), single-cell protein (Hamer, 1979) and polysaccharides (Pace and Righelato, 1980; Smith and Pace, 1982). In the selection of processes for the recovery of biological products it should always be understood that recovery and production are inter-linked, and that good recovery starts in the fermentation by the selection of, amongst other factors, the correct media and time of harvesting.

The recovery and purification of many compounds may be achieved by a number of alternative routes. The decision to follow a particular route involves comparing the following factors to determine the most appropriate under a given set of circumstances:

Capital costs.
Processing costs.
Throughput requirements.
Yield potential.
Product quality.
Technical expertise available.

1. Harvested broth
- ↓
2. Filter off *A. niger* mycelium using a rotary vacuum filter
- ↓
3. Add $\text{Ca}(\text{OH})_2$ to filtrate until pH 5.8
- ↓
4. Calcium citrate
- ↓
5. Add H_2SO_4 while at 60°C
- ↓
6. Filter on rotary vacuum filter to recover CaSO_4
- ↓
7. Activated charcoal to decolourise
- ↓
8. Cation and anion exchange resins
- ↓
9. Evaporate to point of crystallization at 36°C
- ↓
10. Crystals of citric monohydrate separated in continuous centrifuges
- ↓
11. Driers at $50-60^\circ\text{C}$

FIG. 10.4. Recovery and purification of citric acid (Sodesk *et al.*, 1981).

1. Supernatant; 400 dm³ pH 8.8
- ↓
2. Dilute; with 400 dm³ demineralised water
- ↓
3. Acidify; with glacial acetic acid to pH 5.2
- ↓
4. Adsorption; add SP-sephadex C25 (~750g), stir 1hr, settling 2hrs
- ↓
5. Collect resin; pack into an adjustable column, remove unadsorbed protein by passing 0.3M ammonium acetate pH 6.0
- ↓
6. Elution; nuclease eluted with 2M ammonium sulphate
- ↓
7. Dialysis; overnight against demineralised water
- ↓
8. Concentration; CH₃ hollow fibre unit (Amicon) mol. wt. cut off 10,000
- ↓
9. Centrifugation; 10,000g for 30 minutes
- ↓
10. Gel filtration; Sephadex G75, 0.01% acetic acid plus 0.1N ammonium acetate
- ↓
11. Freeze drying

FIG. 10.5. Purification of micrococcal nuclease (Darbyshire, 1981).

Conformance to regulatory requirements.
Waste treatment needs.
Continuous or batch processing.
Automation.
Personnel health and safety (Wildfeuer, 1985).

The major problem currently faced in product recovery is the large-scale purification of biologically active molecules. For a process to be economically viable large-scale production is required, and therefore large-scale separation, recovery, and purification. This then requires the transfer of small-scale preparative/analytical technologies (e.g. chromatographic techniques) to the production scale whilst maintaining efficiency of the process, bio-activity of the product and purity of the product so that it conforms with safety legislation and regulatory requirements. Developments in this field, and remaining areas for development are documented by Pyle (1990).

REMOVAL OF MICROBIAL CELLS AND OTHER SOLID MATTER

Microbial cells and other insoluble materials are normally separated from the harvested broth by filtration or centrifugation. Because of the small size of many microbial cells it will be necessary to consider the use of filter aids to improve filtration rates, while heat and flocculation treatments are employed as techniques for increasing sedimentation rates in centrifugation. The methods of cell and cell debris separation described in the following sections have been practised for many years. Bowden *et al.* (1987) review some potential developments in cell recovery. These include the use of electrophoresis and dielectrophoresis to exploit the charged properties of microbial cells, ultrasonic treatment to improve flocculation characteristics and magnetic separations. All these techniques suffer from high cost and scale-up difficulties and currently are not appropriate technologies. Of more current interest is the use of two-phase liquid extraction. Though still most appropriately used for separation of selected soluble components, it is easy to scale up and uses conditions which are gentle on the product.

FOAM SEPARATION

Foam separation depends on using methods which exploit differences in surface activity of materials. The material may be whole cells or molecular such as a protein or colloidal, and is selectively adsorbed or attached to the surface of gas bubbles rising through a liquid, to be concentrated or separated and finally removed by skimming (Fig. 10.6). It may be possible to make some materials surface active by the application

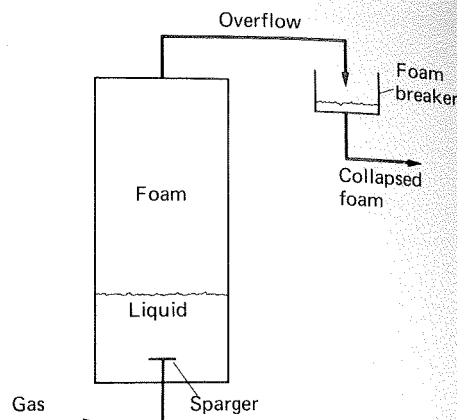


Fig. 10.6. Schematic flow diagram for foam fractionation (Wang and Sinskey, 1970).

of surfactants such as long-chain fatty acids, amines and quaternary ammonium compounds. Materials made surface active and collected are termed colligends whereas the surfactants are termed collectors. When developing this method of separation, the important variables which may need experimental investigation are pH, air-flow rates, surfactants and colligend-collector ratios.

Rubin *et al.* (1966) investigated foam separation of *E. coli* starting with an initial cell concentration of 7.2×10^8 cells cm^{-3} . Using lauric acid, stearyl amine or *t*-octyl amine as surfactants, it was shown that up to 90% of the cells were removed in 1 minute and 99% in 10 minutes. The technique also proved successful with *Chlorella* sp. and *Chlamydomonas* sp. In other work with *E. coli*, Grieves and Wang (1966) were able to achieve cell enrichment ratios of between 10 and 1×10^6 using ethyl-hexadecyl-dimethyl ammonium bromide.

PRECIPITATION

Precipitation may be conducted at various stages of the product recovery process. It is a particularly useful process in that it allows enrichment and concentration in one step, thereby reducing the volume of material for further processing.

It is possible to obtain some products (or to remove certain impurities) directly from the broth by precipitation, or to use the technique after a crude cell lysate has been obtained.

Typical agents used in precipitation render the compound of interest insoluble, and these include:

- (a) Acids and bases to change the pH of a solution until the isoelectric point of the compound is reached and pH equals pI, when there is then no overall charge on the molecule and its solubility is decreased.
- (b) Salts such as ammonium and sodium sulphate are used for the recovery and fractionation of proteins. The salt removes water from the surface of the protein revealing hydrophobic patches which come together causing the protein to precipitate. The most hydrophobic proteins will precipitate first, thus allowing fractionation to take place.
- (c) Organic solvents. Dextrans can be precipitated out of a broth by the addition of methanol. Chilled ethanol and acetone can be used in the precipitation of proteins mainly due to changes in the dielectric properties of the solution.
- (d) Non-ionic polymers such as polyethylene glycol (PEG) can be used in the precipitation of proteins and are similar in behaviour to organic solvents.
- (e) Polyelectrolytes can be used in the precipitation of a range of compounds, in addition to their use in cell aggregation.
- (f) Protein binding dyes (triazine dyes) bind to and precipitate certain classes of protein (Lowe and Stead, 1985).
- (g) Affinity precipitants are an area of much current interest in that they are able to bind to, and precipitate, compounds selectively (Niederauer and Glatz, 1992).

FILTRATION

Filtration is one of the most common processes used at all scales of operation to separate suspended particles from a liquid or gas, using a porous medium which retains the particles but allows the liquid or gas to pass through. Gas filtration has been discussed in detail elsewhere (Chapters 5 and 7). It is possible to carry out filtration under a variety of conditions, but a number of factors will obviously influence the choice of the most suitable type of equipment to meet the specified requirements at minimum overall cost, including:

1. The properties of the filtrate, particularly its viscosity and density.

2. The nature of the solid particles, particularly their size and shape, the size distribution and packing characteristics.
3. The solids : liquid ratio.
4. The need for recovery of the solid or liquid fraction or both.
5. The scale of operation.
6. The need for batch or continuous operation.
7. The need for aseptic conditions.
8. The need for pressure or vacuum suction to ensure an adequate flow rate of the liquid.

Theory of filtration

A simple filtration apparatus is illustrated in Fig. 10.7, which consists of a support covered with a porous filter cloth. A filter cake gradually builds up as filtrate passes through the filter cloth. As the filter cake increases in thickness the resistance to flow will gradually increase. Thus, if the pressure applied to the surface of the slurry is kept constant the rate of flow will gradually diminish. Alternatively, if the flow rate is to be kept constant the pressure will gradually have to be increased. The flow rate may also be reduced by blocking of holes in the filter cloth and closure of voids between particles, if the particles are soft and compressible. When particles are compressible it may not be feasible to apply increased pressure.

Flow through a uniform and constant depth porous bed can be represented by the Darcy equation:

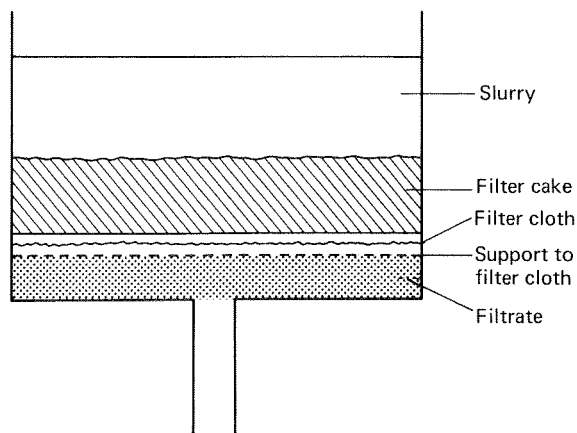


FIG. 10.7. Diagram of a simple filtration apparatus.

$$\text{Rate of flow} = \frac{dV}{dt} = \frac{KA\Delta P}{\mu L} \quad (10.1)$$

where μ = liquid viscosity,
 L = depth of the filter bed,
 ΔP = pressure differential across the filter bed,
 A = area of the filter exposed to the liquid,
 K = constant for the system.

K itself is a term which depends on the specific surface area s (surface area/unit volume) of the particles making up the filter bed and the voidage Σ when they are packed together. The voidage is the amount of filter-bed area which is free for the filtrate to pass through. It is normally 0.3 to 0.6 of the cross-sectional area of the filter bed. Thus K (Kozeny's constant) can be expressed as

$$K = \frac{\Sigma^2}{5(1 - \Sigma)^2 s^2}.$$

Unfortunately, s and Σ are not easily determined.

In most practical cases L is not readily measured but can be defined in terms of:

V = volume of filtrate passed in time t and

ν = volume of cake deposited per unit volume of filtrate.

Then
$$L = \frac{\nu V}{A}.$$

Substituting in equation (10.1):

$$\frac{dV}{dt} = \frac{KA^2\Delta P}{\mu\nu V}. \quad (10.2)$$

This is a general equation relating rate of filtration to pressure drop, cross-sectional area of the filter and filtrate retained. Equation 10.2 can be integrated for filtration at constant pressure.

$$VdV = \frac{KA^2\Delta P}{\mu\nu} dt \quad (10.3)$$

Integrating equation (10.3):

$$V^2 = \frac{2KA^2\Delta P t}{\mu\nu}. \quad (10.4)$$

Now in equation (10.4), ΔP is constant, μ is generally equal to 1, ν can be determined by laboratory investigation and A^2 remains approximately constant. Thus there is a linear relationship between V^2 and t . By carrying out small-scale filtration trials it is therefore possible to obtain a value for K . It is then possible to reapply the equation for large-scale filtration calculations.

Although it is also possible to derive the equation

for the pressure necessary to maintain a constant filtration rate, it has little practical application. The pressure is made up of two components. Firstly the pressure needed to pass the constant volume through the filter resistance and, secondly, an increasing pressure component which is proportional to the resistance from the increasing cake depth. This filtration procedure would be complex to perform practically, and other methods of filtration are used to achieve constant flow rates, e.g. vacuum drum filters.

The use of filter aids

It is common practice to use filter aids when filtering bacteria or other fine or gelatinous suspensions which prove slow to filter or partially block a filter. Kieselguhr (diatomaceous earth) is the most widely used material. It has a voidage of approximately 0.85, and, when it is mixed with the initial cell suspension, improves the porosity of a resulting filter cake leading to a faster flow rate. Alternatively, it may be used as an initial bridging agent in the wider pores of a filter to prevent or reduce blinding. The term 'blinding' means the wedging of particles which are not quite large enough to pass through the pores, so that an appreciable fraction of the filter surface becomes inactive.

The minimum quantity of filter aid to be used in filtration of a broth should be established experimentally. Kieselguhr is not cheap, and it will also absorb some of the filtrate, which will be lost when the filter cake is disposed. The main methods of using the filter aid are:

1. A thin layer of kieselguhr is applied to the filter to form a precoat prior to broth filtration.
2. The appropriate quantity of filter aid is mixed with the harvested broth. Filtration is started, to build up a satisfactory filter bed. The initial raffinate is returned to the remaining broth prior to starting the true filtration.
3. When vacuum drum filters are to be used which are fitted with advancing knife blades, a thick precoat filter is initially built up on the drum (later section in this chapter).

In some processes such as microbial biomass production, filter aids cannot be used and cell pretreatment by flocculation or heating must be considered (see later section in this chapter). In addition it is not normally practical to use filter aids when the product is intracellular and its removal would present a further stage of purification.

Batch filters

PLATE AND FRAME FILTERS

A plate and frame filter is a pressure filter in which the simplest form consists of plates and frames arranged alternately. The plates are covered with filter cloths (Fig. 10.8) or filter pads. The plates and frames are assembled on a horizontal framework and held together by means of a hand screw or hydraulic ram so that there is no leakage between the plates and frames which form a series of liquid-tight compartments. The slurry is fed to the filter frame through the continuous channel formed by the holes in the corners of the plates and frames. The filtrate passes through the filter cloth or pad, runs down grooves in the filter plates and is then discharged through outlet taps to a channel. Sometimes, if aseptic conditions are required, the outlets may lead directly into a pipe. The solids are retained within the frame and filtration is stopped when the frames are completely filled or when the flow of filtrate becomes uneconomically low.

On an industrial scale the plate and frame filter is one of the cheapest filters per unit of filtering space and requires the least floor space, but it is intermittent in operation (a batch process) and there may be considerable wear of filter cloths as a result of frequent dismantling. This type of filter is most suitable for fermentation broths with a low solids content and low resistance to filtration. It is widely used as a 'polishing' device in breweries to filter out residual yeast cells

following initial clarification by centrifugation or rotary vacuum filtration. It may also be used for collecting high value solids that would not justify the use of a continuous filter. Because of high labour costs and the time involved in dismantling, cleaning and reassembly, these filters should not be used when removing large quantities of worthless solids from a broth.

PRESSURE LEAF FILTERS

There are a number of intermittent batch filters usually called by their trade names. These filters incorporate a number of leaves, each consisting of a metal framework of grooved plates which is covered with a fine wire mesh, or occasionally a filter cloth and often precoated with a layer of cellulose fibres. The process slurry is fed into the filter which is operated under pressure or by suction with a vacuum pump. Because the filters are totally enclosed it is possible to sterilize them with steam. This type of filter is particularly suitable for 'polishing' large volumes of liquids with low solids content or small batch filtrations of valuable solids.

(i) Vertical metal-leaf filter

This filter consists of a number of vertical porous metal leaves mounted on a hollow shaft in a cylindrical pressure vessel. The solids from the slurry gradually build up on the surface of the leaves and the filtrate is removed from the plates via the horizontal hollow shaft. In some designs the hollow shaft can be slowly rotated during filtration. Solids are normally removed at the end of a cycle by blowing air through the shaft and into the filter leaves.

(ii) Horizontal metal-leaf filter

In this filter the metal leaves are mounted on a vertical hollow shaft within a pressure vessel. Often, only the upper surfaces of the leaves are porous. Filtration is continued until the cake fills the space between the disc-shaped leaves or when the operational pressure has become excessive. At the end of a process cycle, the solid cake can be discharged by releasing the pressure and spinning the shaft with a drive motor.

(iii) Stacked-disc filter

One kind of filter of this type is the Metafilter. This is a very robust device and because there is no filter cloth and the bed is easily replaced, labour costs are low. It consists of a number of precision-made rings which are stacked on a fluted rod (Fig. 10.9). The rings

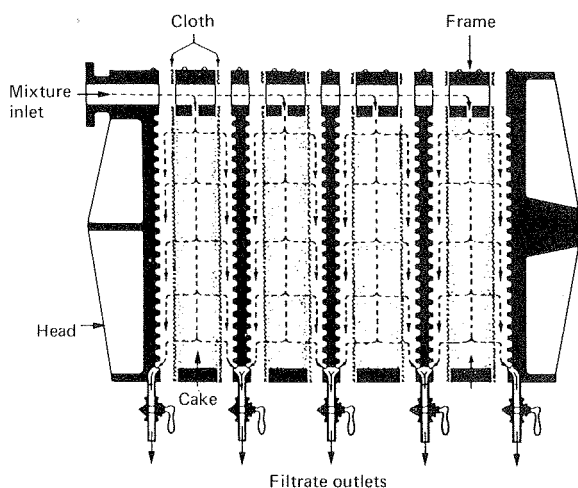


FIG. 10.8. Flush plate and frame filter assembly. The cloth is shown away from the plates to indicate flow of filtrate in the grooves between pyramids (Purchas, 1971).

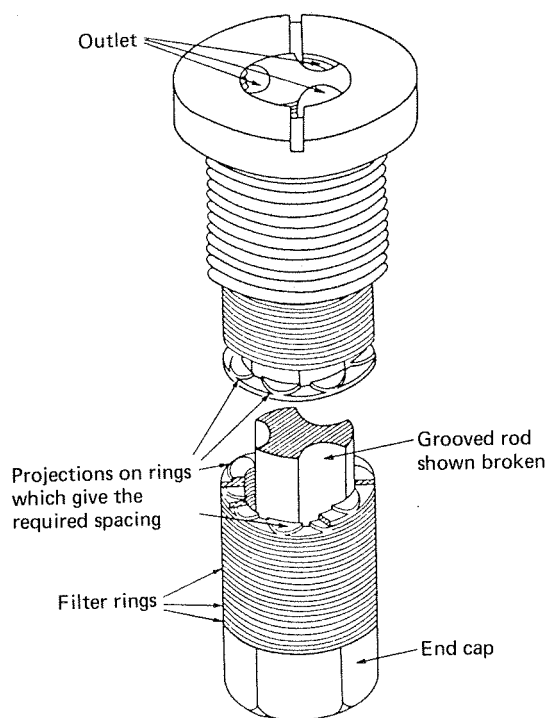


FIG. 10.9a. Metafilter pack (Coulson and Richardson, 1991).

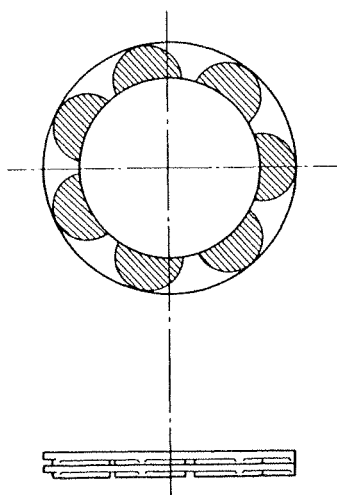


FIG. 10.9b. Rings for metafilter (Coulson and Richardson, 1991).

(22 mm external diameter, 16 mm internal diameter and 0.8 mm thick) are normally made from stainless steel and precision stamped so that there are a number of shoulders on one side. This ensures that there will be clearances of 0.025 mm to 0.25 mm when the rings

are assembled on the rods. The assembled stacks are placed in a pressure vessel which can be sterilized if necessary. The packs are normally coated with a thin layer of kieselguhr which is used as a filter aid. During use, the filtrate passes between the discs and is removed through the grooves of the fluted rods, while solids are deposited on the filter coating. Operation is continued until the resistance becomes too high and the solids are removed from the rings by applying back pressure via the fluted rods. Metafilters are primarily used for 'polishing' liquids such as beer.

Continuous filters

ROTARY VACUUM FILTERS

Large rotary vacuum filters are commonly used by industries which produce large volumes of liquid which need continuous processing. The filter consists of a rotating, hollow, segmented drum covered with a fabric or metal filter which is partially immersed in a trough containing the broth to be filtered (Fig. 10.10). The slurry is fed on to the outside of the revolving drum and vacuum pressure is applied internally so that the filtrate is drawn through the filter, into the drum and finally to a collecting vessel. The interior of the drum is divided into a series of compartments, to which the vacuum pressure is normally applied for most of each revolution as the drum slowly revolves (~ 1 rpm). However, just before discharge of the filter cake, air pressure may be applied internally to help ease the filter cake off the drum. A number of spray jets may be carefully positioned so that water can be applied to rinse the cake. This washing is carefully controlled so that dilution of the filtrate is minimal.

It should be noted that the driving force for filtration (pressure differential across the filter) is limited to one atmosphere (100 kN m^{-2}) and in practice it is significantly less than this. In contrast, pressure filters can be operated at many atmospheres pressure. A number of rotary vacuum drum filters are manufactured, which differ in the mechanism of cake discharge from the drum:

- (i) String discharge.
- (ii) Scraper discharge.
- (iii) Scraper discharge with precoating of the drum.

(i) String discharge

Fungal mycelia produce a fibrous filter cake which can easily be separated from the drum by string discharge (Fig. 10.11). Long lengths of string 1.5 cm apart

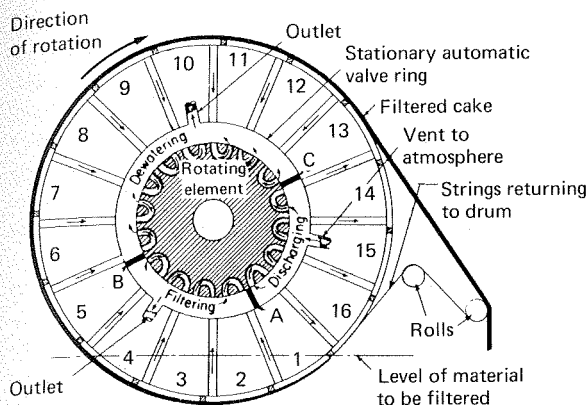


FIG. 10.10. Diagram of string-discharge filter operation. Sections 1 to 4 are filtering; sections 5 to 12 are dewatering; and section 13 is discharging the cake with the string discharge. Sections 14, 15 and 16 are ready to start a new cycle. A, B and C represent dividing members in the annular ring (Miller *et al.*, 1973).

are threaded over the drum and round two rollers. The cake is lifted free from the upper part of the drum when the vacuum pressure is released and carried to the small rollers where it falls free.

(ii) Scraper discharge

Yeast cells can be collected on a filter drum with a knife blade for scraper discharge (Fig. 10.12). The filter cake which builds up on the drum is removed by an accurately positioned knife blade. Because the knife is close to the drum, there may be gradual wearing of the filter cloth on the drum.

(iii) Scraper discharge with precoating of the drum

The filter cloth on the drum can be blocked by bacterial cells or mycelia of actinomycetes. This prob-

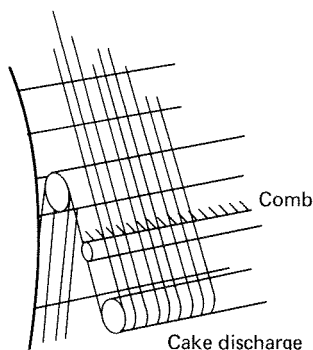


FIG. 10.11. Cake discharge on a drum filter using strings (Talcott *et al.*, 1980).

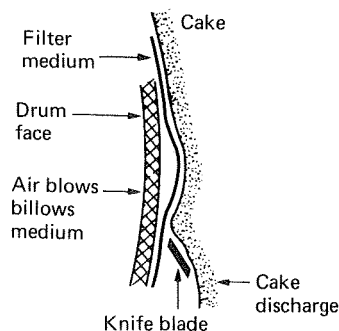


FIG. 10.12. Cake discharge on a drum using a scraper (Talcott *et al.*, 1980).

lem is overcome by precoating the drum with a layer of filter-aid 2 to 10 cm thick. The cake which builds up on the drum during operation is cut away by the knife blade (Fig. 10.13) which mechanically advances towards the drum at a controlled slow rate. Alternatively, the blade may be operated manually when there is an indication of 'blinding' which may be apparent from a reduction in the filtration rate. In either case the cake is removed together with a very thin layer of precoat. A study of precoat drum filtration has been made by Bell and Hutto (1958). The operating variables studied included drum speed, extent of drum submergence, knife advance speed and applied vacuum. The work indicated that optimization for a new process might require prolonged trials. Although primarily used for the separation of micro-organisms from broth, studies have indicated (Gray *et al.*, 1973) that rotary vacuum filters can be effective in the processing of disrupted cells.

Cross-flow filtration (tangential filtration)

In the filtration processes previously described, the flow of broth was perpendicular to the filtration mem-

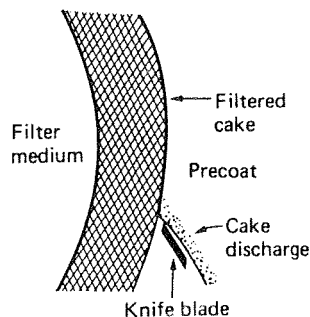


FIG. 10.13. Cake discharge on a precoated drum filter (Talcott *et al.*, 1980).